TM Forum Specification

Usage Consumption User Guide

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Introduction

The following document is the specification of the REST API for Usage Consumption. It includes the model definition as well as all available operations.

This API covers the consumption follow up function providing ongoing information about usages related to any subscribed network products (voice, data, TV,…) without having to wait for the invoice/bill generation. This information may concern usages charged against a bucket (supervised or not) and the remaining balances of the bucket. It allows customers or users to be informed of usages and remaining credits on the buckets that they consume under their purchased offers and options.

A bucket is a quantity of units (time, volume, currency or events, e.g. SMS) available via a subscribed offer or option and that can be consumed during a validity period. A bucket has at least one balance valid at a given point in time. This balance is decremented by usage charged against the bucket and incremented by credit operations (like top-up operations for example). At a given point, this balance defines the remaining allowed usage quantity for a given period, corresponding to the initial allowed usage quantity minus consumed usage quantity.

The bucket has also consumption counters which give measures about the usage quantity consumed on the bucket. These counters can be detailed by device or by user for example when a bucket is shared between several devices or several users.

The usage consumption API allow to view at a given point the balance and the consumption counters of the various buckets (SMS, Voice, Data for example) that one or more user(s) consume with each of his devices, according to the purchased offers and options.

A usage consumption report retrieves the data related to these balances and various consumption counters and calculated at the time of the request by the server.

The API allows to retrieve usage consumption about balances and consumption counters for a given criteria : a network product or a user for example.

The use of the both resources described in this document is conditioned by the capability of the server.

- If the server could perform the calculation and provide the usage consumption report in synchronous mode according to the criteria given by the API requester, a GET operation with filter criteria on usageConsumptionReport resource could be used. We could accept to use a GET operation in this context because the server calculates the data given in the usage consumption report without necessarily store it.
- We must also anticipate the use case where the server could only give an asynchronous response to the calculation of an usage consumption report requested by the API requester. In this case, a POST operation on queryUsageConsumption could be sent by the API Requester asking the server that a usage consumption report calculation has to be performed. Subsequently, the API requester could follow the process with a GET operation on this same resource retrieving the status of the request. The server could also use the notification pattern to inform the API requester that the usage consumption report is available.

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It describes the following managed resources, called QueryUsageConsumption and UsageConsumptionReport.

QueryUsageConsumption is a Task Resource.

QueryUsageConsumption covers the consumption levels of products related to one or more parties. We can search for the remaining, global or used bucket values, or for exceptional consumption normally termed as out of bucket consumption or overage.

Usage Consumption Buckets may be linked between each other (aggregated buckets, bundled buckets, etc).

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Role based Access Control

The user presents their credentials for authentication

If the credentials are valid

- The user is given access to the usage consumption API
- As defined by their role(s)
- As defined by their access rights
- As defined by the access type: CRUD, discover
- As defined by the pre-defined filter

For example, if they issue a get on a usage consumption report that a party has no access they get an error response.

Normally we anticipate that the OAUTH2 or Open ID Connect are used as the authorization APIs and that ACL are established between authorized parties with regards to the content of the Usage Consumption reports.

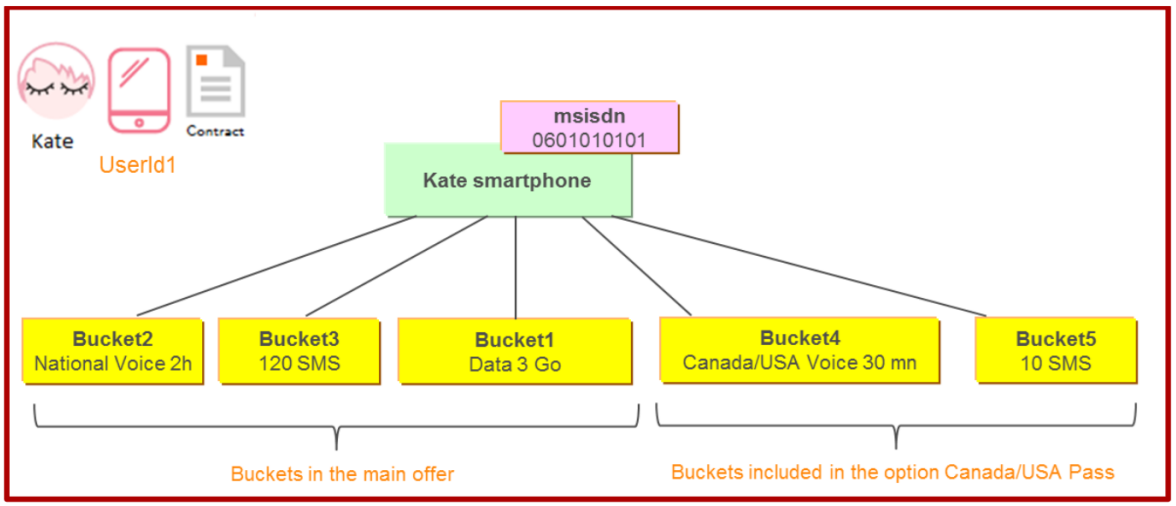
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Sample Use Cases

This section includes a set of main use cases that can be performed with this API. Additional use cases can be generated using the operations and resources defined in this specification.

Use case 1 (UC1) : one user Kate with one device

Kate has a smartphone identified by msisdn number 0601010101. She subscribed to a MainOffer allowing her to consume three monthly buckets : a national voice bucket with an initial value of 2 hours, a SMS bucket with an initial value of 120 SMS and a data bucket with an initial value of 3 Go. She also subscribed to an option CanadaUSAPass with a Canada/USA voice bucket with an initial value of 30 minutes and a SMS bucket with an initial value of 10 SMS.



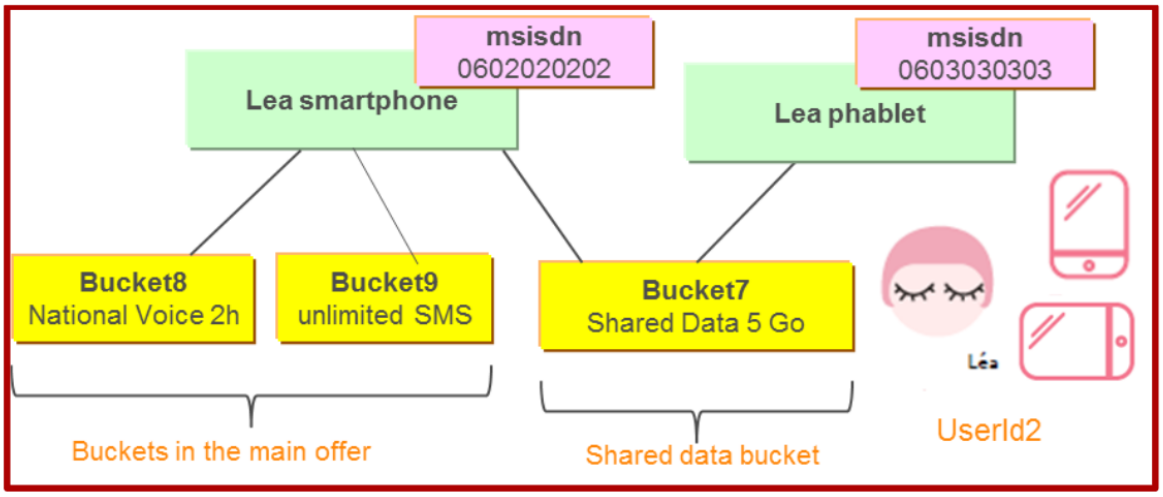
With her smartphone, Kate has used 40 minutes on the national voice bucket, 25 sms on sms bucket and 1.2 Go on the data bucket of the main offer. She also has consumed 20 minutes on the voice bucket and 10 sms ont the sms bucket of the Canada/USA Pass. She also had 20 USD out of bucket in international voice usages.

Use case	Consumer customer opens web portal to check the amount of minutes spent in international calls.
Actors	Customer.
Pre-conditions	None.
Expected outcome	Used amount of national voice minutes displayed to the customer.
Operation	POST /queryUsageConsumption
Description	1. Customer opens web portal, 2. Web portal gets current usage snapshot for customer's products, 3. Web portal presents main tile associated to usage types, 4. Customer clicks on voice tile, 5. Web portal displays remaining and used usage for international and national voice, e.g., 1000 minutes and 5000 minutes.

Use case 2 (UC2) : one user Lea with two devices consuming a shared data bucket

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Lea has two devices : a smartphone identified by msisdn number 0602020202 and a phablet identified by msisdn number 0603030303. She subscribed to a MainOffer allowing her to consume with its smartphone two monthly buckets : a national voice bucket with an initial value of 2 hours and an unlimited SMS bucket. She also subscribed to a Shared Data offer with an initial value of 5 Go. This data bucket is shared with the phablet et can also be consumed by it.



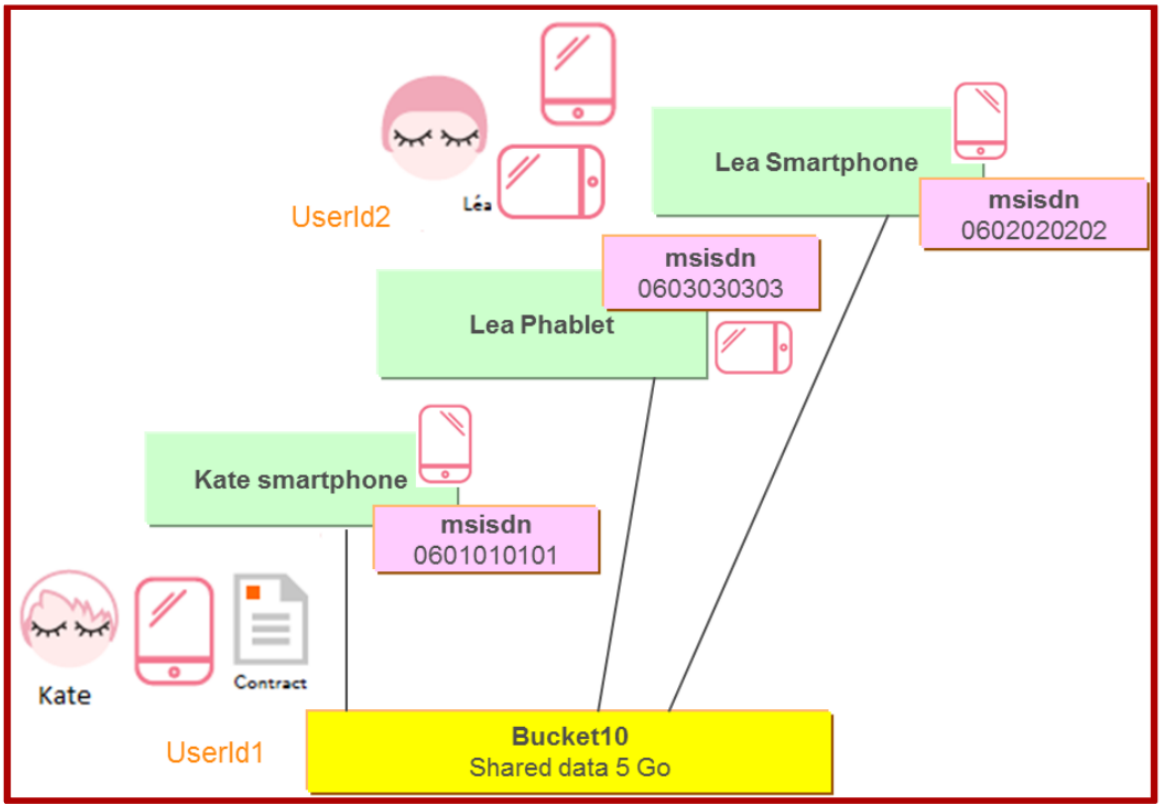
With her smartphone, Lea has used 60 minutes on the voice bucket, 123 sms on the SMS bucket and 1.0 Go on the 3.0 Go used in the data shared bucket. She also has used 2.0 Go with the phablet on the 3.0 Go used in the data shared bucket.

Use case	Consumer customer opens his/her mobile app to check the remaining data's amount.
Actors	Customer.
Pre-conditions	None.
Expected outcome	Remaining amount of data displayed to the customer.
Operation	POST /queryUsageConsumption
Description	1. Customer opens mobile app 2. Mobile app gets current usage snapshot for customer' products 3. Mobile app displays remaining data value

Use case 3 (UC3) : a shared data bucket consumed by a community of two users

Kate is the contract holder. Kate subscribes to a Shared Data offer containing a shared data bucket of 5 Go consumed by a community (family) of two users : Kate and Lea. Kate consumes it with her smartphone identified by msisdn number 0601010101 and Lea with her smartphone identified by msisdn number 0602020202 and her tablet identified by msisdn number 0603030303.

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Lea has used 1 Go with her smartphone and 1.2 Go with the phablet. Kate has used 1.0 Go with her smartphone. For this shared data bucket, we can have :

- A global report for the three products (Kate smartphone, Lea smartphone and Lea phablet) with a global balance (remains 1.8 Go), a global consumption counter (3.2 used), consumption counters detailed by users (1.0 used by Kate and 2.2 used by Lea) and consumption counters detailed by product (1.0 used by Kate smartphone, 1.0 used by Lea smartphone and 1.2 used by Lea phablet).
- A report for a user : Lea for example with consumption counters for her smartphone and her phablet or Kate with consumption counters for her smartphone.
- A report for a product (Kate smartphone, Lea smartphone or Lea phablet).

Use case	Agent validates counters associated to a service.
Actors	Agent
Pre-conditions	None.
Expected outcome	Used amount of data displayed to the customer.
Operation	POST /queryUsageConsumption
Description	1. Agent opens back-office web portal, 2. Agent digits telephone number, 3. Web portal gets current usage snapshot for this service, 4. Web portal displays current counters and balances

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Support of polymorphism and extension patterns

Support of polymorphic collections and types and schema based extension is provided by means of a list of generic meta-attributes that we describe below. Polymorphism in collections occurs when entities inherit from base entities, for instance a BillingAccount and SettlementAccount inheriting properties from the abstract Account entity.

Generic support of polymorphism and pattern extensions is described in the TMF API Guidelines, Part 2 (TMF630).

The @type attribute provides a way to represent the actual class type of an entity. For example, within a list of Account instances some may be instances of BillingAccount where other could be instances of SettlementAccount. The @type gives this information. All resources and sub-resources of this API have a @type attributes that can be provided when this is useful.

The @referredType can be used within reference entities (like for instance an AccountRef object) to explicitly denote the actual entity type of the referred class. Notice that in reference entities the @type, when used, denotes the class type of the reference itself, such as BillingAccountRef or SettlementAccountRef, and not the class type of the referred object. However since reference classes are rarely sub-classed, @type is generally not useful in reference objects.

The @schemaLocation property can be used in resources to allow specifying user-defined properties of an Entity or to specify the expected characteristics of an entity.

The @baseType attribute gives a way to provide explicitly the base of class of a given resource that has been extended.

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RESOURCE MODEL

Managed Entity and Task Resource Models

QueryUsageConsumption resource

An QueryUsageConsumption allows to manage the calculation request of an usage consumption.

Resource model

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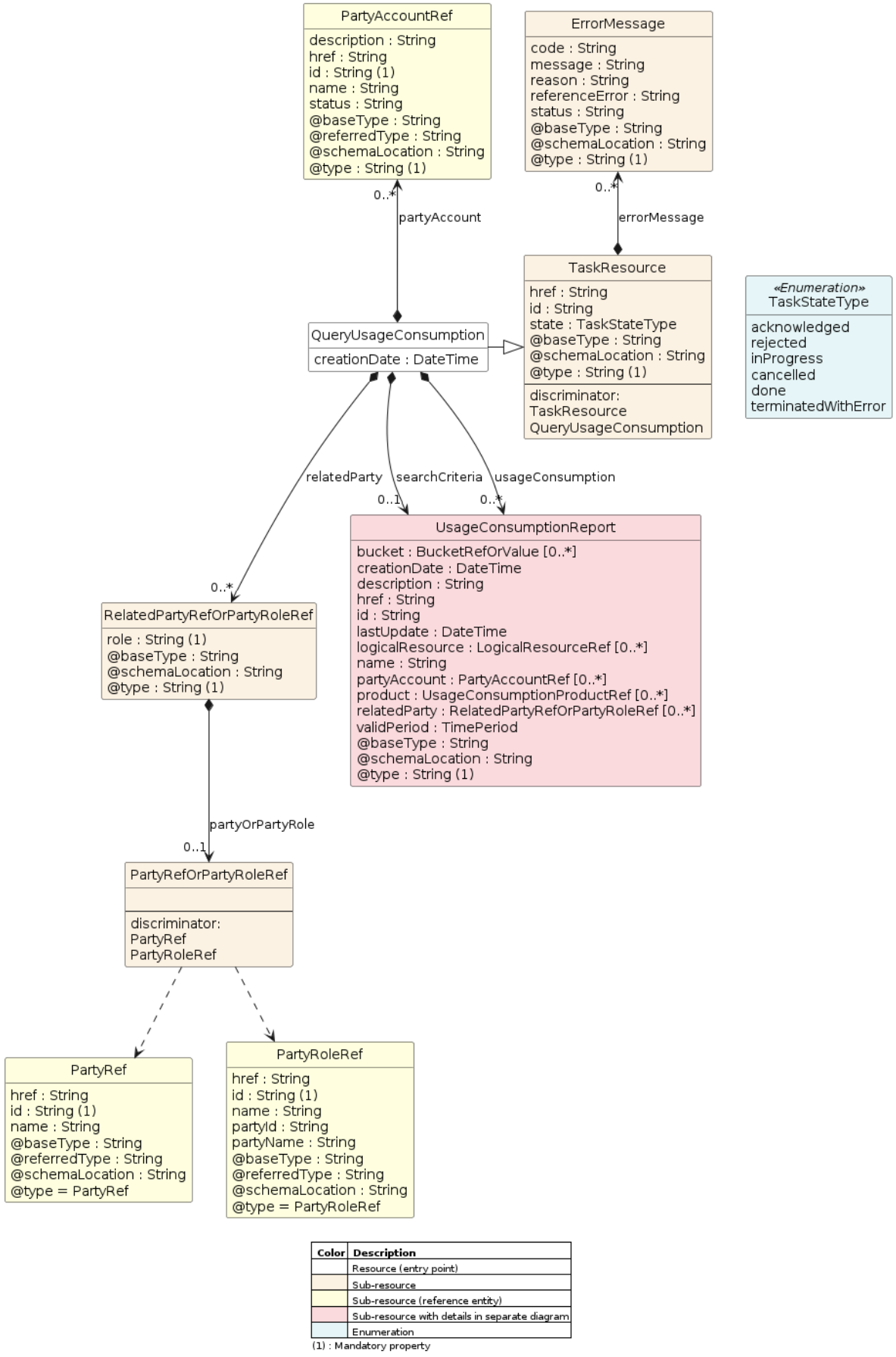


Figure 1 - QueryUsageConsumption

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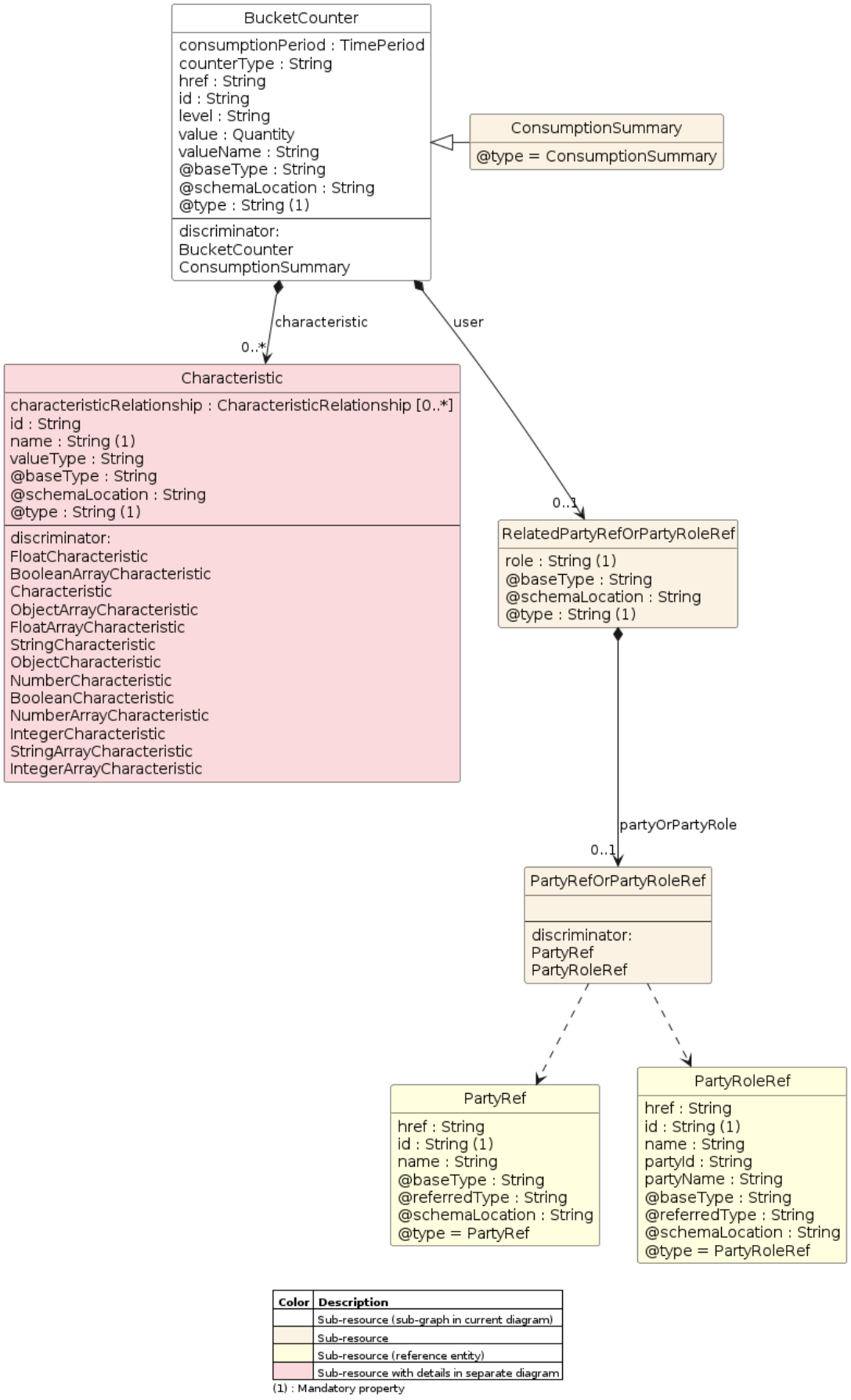


Figure 2 - BucketCounter

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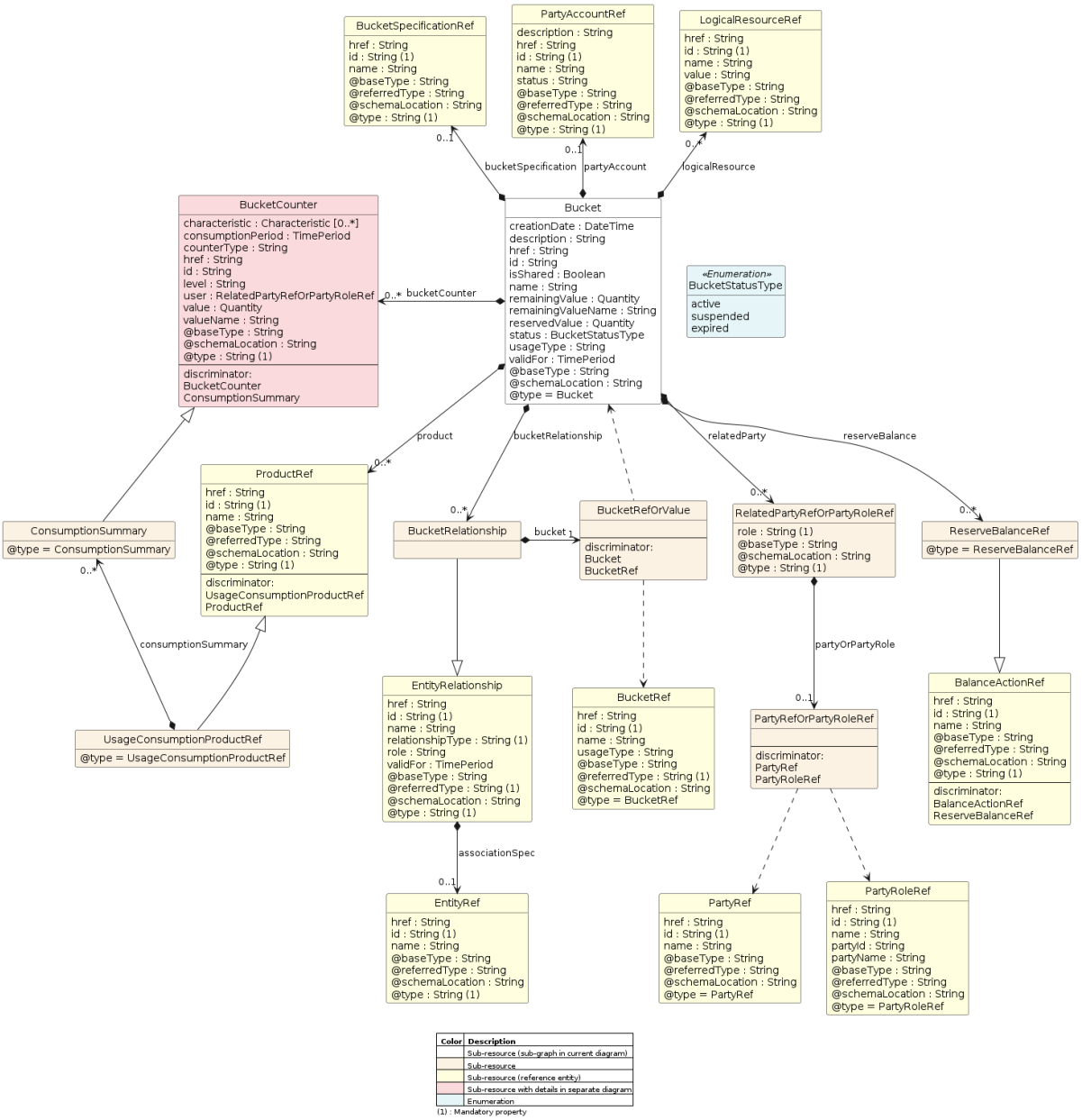


Figure 3 - Bucket

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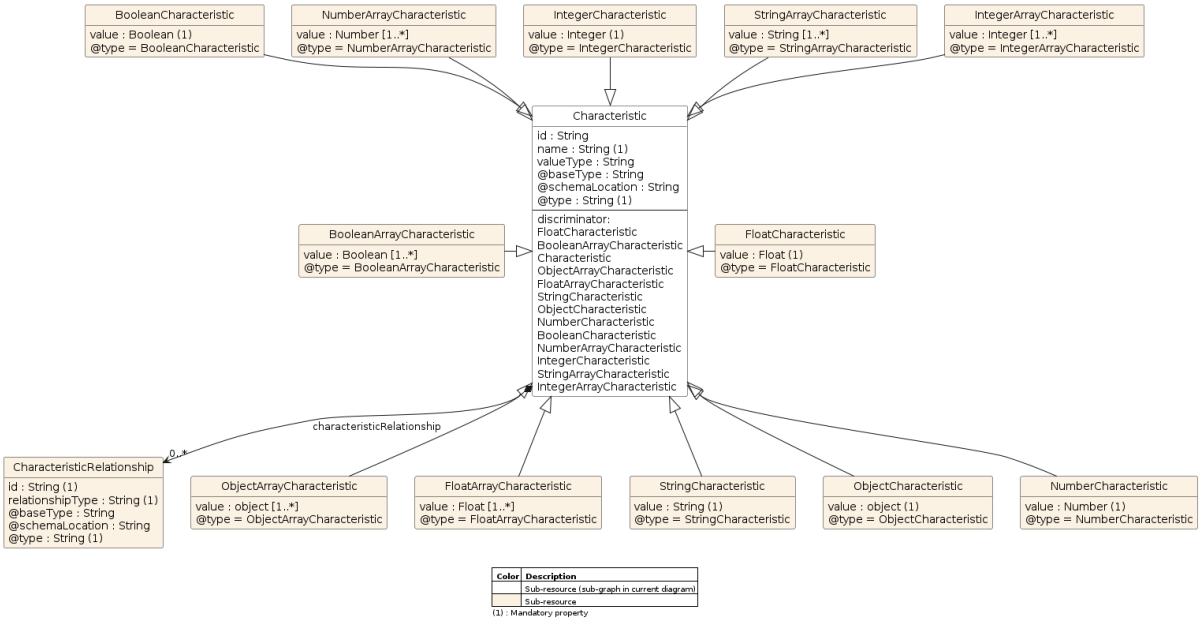


Figure 4 - Characteristic

Field descriptions

QueryUsageConsumption fields

creationDate	A DateTime. Date when the query was submitted.
errorMessage	An ErrorMessage. A list of error message contains the error that caused the termination of the task.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
partyAccount	A PartyAccountRef. A reference to the account that owns the bucket.
relatedParty	A RelatedPartyRefOrPartyRoleRef. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.
searchCriteria	A UsageConsumptionReport. An usage consumption report enables to know at a given point the balances and the consumption counters related to various buckets (SMS, Voice, Data for example). It could be calculated for a device identified by a public key (msisdn number for a mobile device for example or PSTN or VOIP number for a fix device), for a subscribed offer or option or for an user.
state	A TaskStateType. Possible values for the state of a task. ENUMERATED with values: * acknowledged * rejected * inProgress * cancelled * done * terminatedWithError

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usageConsumption	A UsageConsumptionReport. Report with current counters and balances.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ErrorMessage sub-resource fields

code	A String. Error code.
message	A String. More details and corrective actions related to the error.
reason	A String. Explanation of the reason for the error.
referenceError	A String. URI of documentation describing the error.
status	A String. Error code extension like sys-ABC-2001.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyAccountRef sub-resource fields

description	A String. Detailed description of the party account.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
status	A String. The condition of the account, such as due, paid, in arrears.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedPartyRefOrPartyRoleRef sub-resource fields

partyOrPartyRole	A PartyRefOrPartyRoleRef.

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role	A String. Role played by the related party or party role in the context of the specific entity it is linked to. Such as 'initiator', 'customer', 'salesAgent', 'user'.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRoleRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
partyId	A String. The identifier of the engaged party that is linked to the PartyRole object.
partyName	A String. The name of the engaged party that is linked to the PartyRole object.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

UsageConsumptionReport sub-resource fields

bucket	A BucketRefOrValue. Bucket(s) included in the offer or option subscribed.

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creationDate	A DateTime. Date and time of the request creation.
description	A String. Free short text describing the usage consumption content.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
lastUpdate	A DateTime. Date when the status was last changed.
logicalResource	A LogicalResourceRef. Reference to the LogicalResource eg MSISDN.
name	A String. Usage consumption name.
partyAccount	A PartyAccountRef. A reference to the account that owns the bucket.
product	A UsageConsumptionProductRef.
relatedParty	A RelatedPartyRefOrPartyRoleRef. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.
validPeriod	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Bucket sub-resource fields

bucketCounter	A BucketCounter. The bucket counters (commonly referred to as meters) detail, for example, the different kind of consumptions against a bucket. BucketCounter can be instantiated as * ConsumptionSummary
bucketRelationship	A BucketRelationship. Linked bucket to the one instantiate, such as [isAggregated] if the bucket is a part of other or [aggregator] is the bucket holds others.
bucketSpecification	A BucketSpecificationRef. Bucket specification reference. A bucket specification gives additional details on the bucket in a given context, e.g. voice minutes, data, prepaid balance, digital wallet.
creationDate	A DateTime. Date and time when the product was created.
description	A String. Text describing the contents of the balance managed by the bucket.
href	A String. Hyperlink reference.
id	A String. Unique identifier.

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isShared	A Boolean. True if the bucket is shared between several devices or users.
logicalResource	A LogicalResourceRef. Reference to the LogicalResource eg MSISDN.
name	A String. Friendly name to identify the bucket.
partyAccount	A PartyAccountRef. PartyAccount reference. A party account is an arrangement that a party has with an enterprise that provides products to the party.
product	A ProductRef. A Product reference. ProductRef can be instantiated as * UsageConsumptionProductRef
relatedParty	A RelatedPartyRefOrPartyRoleRef. Used to provide information about any other entity with relation to the balance, for instance to define customer hierarchy for the balance (e.g.: customerId, userId,).
remainingValue	A Quantity. An amount in a given unit.
remainingValueName	A String. Remaining amount in a formatted string for the bucket given in the balance unit (for example 1.9 Gb). This formatted string could be used for display needs for example.
reserveBalance	A ReserveBalanceRef. The ReserveBalance resource allows adjustments to be made to the original BalanceTopup eg increment the amount, alter the recharge periodicity.
reservedValue	A Quantity. An amount in a given unit.
status	A BucketStatusType. Possible values for the bucket status are (for example active,suspended,expired). ENUMERATED with values: * active * suspended * expired
usageType	A String. Defines the type of the underlying Balance eg data,voice, any currency eg EUR, USD etc.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BucketCounter sub-resource fields

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characteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic… Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
consumptionPeriod	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
counterType	A String. Type of the consumption counter. We can give for example a counter of the used value for a bucket (counterType=used for example) or the value of the consumption done out of the bucket(s) (counterType=outOfBucket for example).
href	A String. Hyperlink reference.
id	A String. Unique identifier.
level	A String. Counter level. The counter can be given globally for the bucket or detailed by user or by network product for example in case of shared bucket.
user	A RelatedPartyRefOrPartyRoleRef. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.
value	A Quantity. An amount in a given unit.
valueName	A String. Value of the counter in a formatted string used for display needs for example.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	BucketCounter can be instantiated as * ConsumptionSummary

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Characteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
value	This property is present in subclasses

CharacteristicRelationship sub-resource fields

id	A String. Unique identifier of the characteristic.
relationshipType	A String. The type of relationship.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BooleanArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Boolean. A characteristic which value is an array of Boolean(s).

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valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BooleanCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Boolean. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

FloatArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Float. A characteristic which value is an array of Float(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

FloatCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.

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value	A Float. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntegerArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An Integer. A characteristic which value is an array of Integer(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntegerCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An Integer. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

NumberArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.

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name	A String. Name of the characteristic.
value	A Number. A characteristic which value is an array of Number(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

NumberCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Number. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ObjectArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An object. Collection of characteristic values.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ObjectCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.

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id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An object. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StringArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A String. Collection of string characteristics.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StringCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A String. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ConsumptionSummary sub-resource fields

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characteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic… Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
consumptionPeriod	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
counterType	A String. Type of the consumption counter. We can give for example a counter of the used value for a bucket (counterType=used for example) or the value of the consumption done out of the bucket(s) (counterType=outOfBucket for example).
href	A String. Hyperlink reference.
id	A String. Unique identifier.
level	A String. Counter level. The counter can be given globally for the bucket or detailed by user or by network product for example in case of shared bucket.
user	A RelatedPartyRefOrPartyRoleRef. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.
value	A Quantity. An amount in a given unit.
valueName	A String. Value of the counter in a formatted string used for display needs for example.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BucketRelationship sub-resource fields

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associationSpec	An EntityRef. Entity reference schema to be use for all entityRef class.
bucket	A BucketRefOrValue. A bucket (called UsageVolumProduct in the SID Model) represents and tracks a quantity of usage (remaining or consumed). It could be either a quantity or an amount in a currency. It can represent a fixed number of SMS, MMS, call minutes, quantity of data and so on.
href	A String.
id	A String.
name	A String.
relationshipType	A String. Type of relationship such as migration, substitution, dependency, exclusivity.
role	A String. The association role for this entity.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String.

EntityRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

EntityRelationship sub-resource fields

associationSpec	An EntityRef. Entity reference schema to be use for all entityRef class.
href	A String.
id	A String.
name	A String.

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relationshipType	A String. Type of relationship such as migration, substitution, dependency, exclusivity.
role	A String. The association role for this entity.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String.
	EntityRelationship can be instantiated as * BucketRelationship
bucket	This property is present in subclasses

BucketSpecificationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

LogicalResourceRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
value	A String. Value used to identifiy this resource as a MSISDN or a VoIP number.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

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ProductRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	ProductRef can be instantiated as * UsageConsumptionProductRef
consumptionSummary	This property is present in subclasses

UsageConsumptionProductRef sub-resource fields

consumptionSummary	A ConsumptionSummary. The consumption counters (called ConsumptionSummary in the SID model) detail, for example, the different kinds of usage consumption.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ReserveBalanceRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String. When sub-classing, this defines the sub-class Extensible name.

BalanceActionRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	BalanceActionRef can be instantiated as * ReserveBalanceRef

BucketRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
usageType	A String. Defines the type of the underlying Balance eg data,voice, any currency eg EUR, USD etc.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

LogicalResourceRef sub-resource fields

See [LogicalResourceRef] for details.

UsageConsumptionProductRef sub-resource fields

See [UsageConsumptionProductRef] for details.

ProductRef sub-resource fields

See [ProductRef] for details.

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UsageConsumptionReport sub-resource fields

See [UsageConsumptionReport] for details.

Json representation sample(s)

The following example demonstrates the details of the QueryUsageConsumption resource.


```
{
  "id": "7655",
  "href": "https://mycsp.com:8080/tmf-
api/usageConsumption/v5/queryUsageConsumption/7655",
  "relatedParty": [
    {
      "role": "Agent",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "id": "3426",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/3426",
        "name": "Bob Smith",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    }
  ],
  "searchCriteria": {
    "bucket": [
      {
        "usageType": "voice national",
        "@type": "Bucket"
      },
      {
        "usageType": "voice international",
        "@type": "Bucket"
      },
      {
        "usageType": "data",
        "@type": "Bucket"
      }
    ],
    "@type": "UsageConsumptionReport"
  },
  "usageConsumption": [
    {
      "id": "UC1",
      "creationDate": "2021-06-18T16:04:09.084Z",
      "lastUpdate": "2021-06-18T16:04:09.084Z",
      "bucket": [
        {
          "id": "BK1",
          "href": "https://mycsp.com:8080/tmf-api/balanceManagement/v5/bucket/BK1",
          "description": "Bucket related to international calls",
          "isShared": false,
          "name": "RED Voice Plus Bucket - International calls",
          "remainingValue": {
            "amount": 2000,
            "units": "minutes"
          },
          "remainingValueName": "2000 minutes",
          "usageType": "voice international",
```


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```
          "bucketCounter": [
            {
              "id": "CT1",
              "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/bucketCounter/CT1",
              "counterType": "used",
              "level": "global",
              "consumptionPeriod": {
                "endDateTime": "2021-06-01T16:04:09.084Z",
                "startDateTime": "2021-06-18T16:04:09.084Z"
              },
              "value": {
                "amount": 1000,
                "units": "minutes"
              },
              "@type": "BucketCounter"
            }
          ],
          "bucketRelationship": [
            {
              "relationshipType": "aggregated",
              "bucket": {
                "id": "BK2",
                "href": "https://mycsp.com:8080/tmf-
api/usageConsumption/v5/bucket/BK2",
                "@type": "BucketRef"
              },
              "@type": "BucketRelationship"
            }
          ],
          "product": [
            {
              "id": "PR1",
              "href": "https://mycsp.com:8080/tmf-
api/productInventory/v5/product/PR1",
              "name": "RED Voice Plus",
              "@type": "ProductRef"
            }
          ],
          "relatedParty": [
            {
              "role": "Customer",
              "@type": "RelatedPartyRefOrPartyRoleRef",
              "partyOrPartyRole": {
                "id": "3426",
                "href": "https://mycsp.com:8080/tmf-
api/customerManagement/v5/customer/3426",
                "name": "John Doe",
                "@type": "PartyRoleRef",
                "@referredType": "Customer"
              }
            }
          ],
          "@type": "Bucket"
        },
        {
          "id": "BK2",
          "href": "https://mycsp.com:8080/tmf-api/balanceManagement/v5/bucket/BK2",
          "description": "Bucket related to national calls",
          "isShared": false,
          "name": "RED Voice Plus Bucket - National calls",
          "remainingValue": {
            "amount": 4000,
```


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```
            "units": "minutes"
          },
          "remainingValueName": "4000 minutes",
          "usageType": "voice national",
          "bucketCounter": [
            {
              "id": "CT2",
              "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/bucketCounter/CT2",
              "counterType": "used",
              "level": "global",
              "consumptionPeriod": {
                "endDateTime": "2021-06-01T16:04:09.084Z",
                "startDateTime": "2021-06-18T16:04:09.084Z"
              },
              "value": {
                "amount": 1000,
                "units": "minutes"
              },
              "@type": "BucketCounter"
            }
          ],
          "bucketRelationship": [
            {
              "relationshipType": "aggregated",
              "bucket": {
                "id": "BK1",
                "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/bucket/BK2",
                "@type": "BucketRef"
              },
              "@type": "BucketRelationship"
            }
          ],
          "product": [
            {
              "id": "PR1",
              "href": "https://mycsp.com:8080/tmf-
api/productInventory/v5/product/PR1",
              "name": "RED Voice Plus",
              "@type": "ProductRef"
            }
          ],
          "relatedParty": [
            {
              "role": "Customer",
              "@type": "RelatedPartyRefOrPartyRoleRef",
              "partyOrPartyRole": {
                "id": "3426",
                "href": "https://mycsp.com:8080/tmf-
api/customerManagement/v5/customer/3426",
                "name": "John Doe",
                "@type": "PartyRoleRef",
                "@referredType": "Customer"
              }
            }
          ],
          "@type": "Bucket"
        },
        {
          "id": "BK3",
          "href": "https://mycsp.com:8080/tmf-api/balanceManagement/v5/bucket/BK3",
          "description": "Bucket related to data",
```


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```
          "isShared": true,
          "name": "RED Shared Data 10 - Data",
          "usageType": "data",
          "bucketCounter": [
            {
              "id": "CT3",
              "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/bucketCounter/CT3",
              "counterType": "used",
              "level": "detailed by user",
              "consumptionPeriod": {
                "endDateTime": "2021-06-01T16:04:09.084Z",
                "startDateTime": "2021-06-18T16:04:09.084Z"
              },
              "user": {
                "id": "CUST2",
                "name": "Helen Water",
                "role": "Shared data member",
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "@referredType": "Individual"
              },
              "value": {
                "amount": 1,
                "units": "GB"
              },
              "@type": "BucketCounter"
            },
            {
              "id": "CT4",
              "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/bucketCounter/CT4",
              "counterType": "used",
              "level": "detailed by user",
              "consumptionPeriod": {
                "endDateTime": "2021-06-01T16:04:09.084Z",
                "startDateTime": "2021-06-18T16:04:09.084Z"
              },
              "user": {
                "id": "CUST3",
                "name": "Shawn Water",
                "role": "Shared data member",
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "@referredType": "Individual"
              },
              "value": {
                "amount": 0.7,
                "units": "GB"
              },
              "@type": "BucketCounter"
            },
            {
              "id": "CT5",
              "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/bucketCounter/CT5",
              "counterType": "used",
              "level": "detailed by user",
              "consumptionPeriod": {
                "endDateTime": "2021-06-01T16:04:09.084Z",
                "startDateTime": "2021-06-18T16:04:09.084Z"
              },
              "user": {
                "id": "CUST1",
                "name": "John Water",
```


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```
                "role": "Customer",
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "@referredType": "Individual"
              },
              "value": {
                "amount": 1.5,
                "units": "GB"
              },
              "@type": "BucketCounter"
            },
            {
              "id": "CT6",
              "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/bucketCounter/CT6",
              "counterType": "remaining",
              "level": "detailed by user",
              "consumptionPeriod": {
                "endDateTime": "2021-06-01T16:04:09.084Z",
                "startDateTime": "2021-06-18T16:04:09.084Z"
              },
              "user": {
                "id": "CUST2",
                "name": "Helen Water",
                "role": "Shared data member",
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "@referredType": "Individual"
              },
              "value": {
                "amount": 1,
                "units": "GB"
              },
              "@type": "BucketCounter"
            },
            {
              "id": "CT7",
              "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/bucketCounter/CT7",
              "counterType": "remaining",
              "level": "detailed by user",
              "consumptionPeriod": {
                "endDateTime": "2021-06-01T16:04:09.084Z",
                "startDateTime": "2021-06-18T16:04:09.084Z"
              },
              "user": {
                "id": "CUST3",
                "name": "Shawn Water",
                "role": "Shared data member",
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "@referredType": "Individual"
              },
              "value": {
                "amount": 1.3,
                "units": "GB"
              },
              "@type": "BucketCounter"
            },
            {
              "id": "CT8",
              "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/bucketCounter/CT8",
              "counterType": "remaining",
              "level": "detailed by user",
              "consumptionPeriod": {
```


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```
                "endDateTime": "2021-06-01T16:04:09.084Z",
                "startDateTime": "2021-06-18T16:04:09.084Z"
              },
              "user": {
                "id": "CUST1",
                "name": "John Water",
                "role": "Customer",
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "@referredType": "Individual"
              },
              "value": {
                "amount": 4.5,
                "units": "GB"
              },
              "@type": "BucketCounter"
            }
          ],
          "product": [
            {
              "id": "PR2",
              "href": "https://mycsp.com:8080/tmf-
api/productInventory/v5/product/PR2",
              "name": "RED Shared Data 10",
              "@type": "ProductRef"
            }
          ],
          "relatedParty": [
            {
              "role": "Customer",
              "@type": "RelatedPartyRefOrPartyRoleRef",
              "partyOrPartyRole": {
                "id": "3426",
                "href": "https://mycsp.com:8080/tmf-
api/customerManagement/v5/customer/3426",
                "name": "John Doe",
                "@type": "PartyRoleRef",
                "@referredType": "Customer"
              }
            }
          ],
          "@type": "Bucket"
        }
      ],
      "@type": "UsageConsumptionReport"
    }
  ],
  "@type": "QueryUsageConsumption"
}
```


UsageConsumptionReport resource

An usage consumption report enables to know at a given point the balances and the consumption counters related to various buckets (SMS, Voice, Data for example). It could be calculated for a device identified by a public key (msisdn number for a mobile device for example or PSTN or VOIP number for a fix device), for a subscribed offer or option or for an user.

Resource model

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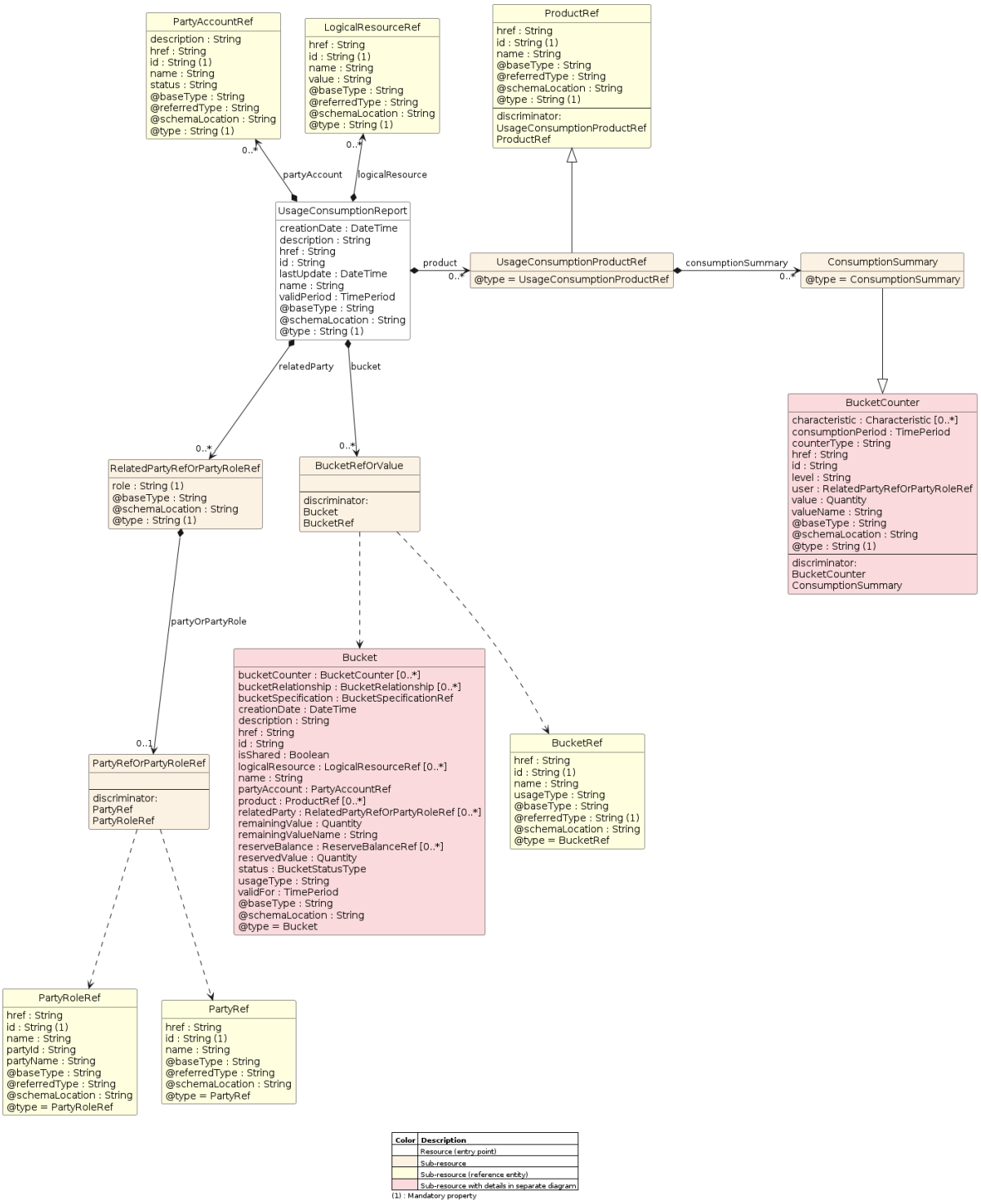


Figure 5 - UsageConsumptionReport

See Figure 2 for BucketCounter diagrams.

See Figure 3 for Bucket diagrams.

See Figure 4 for Characteristic diagrams.

Field descriptions

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UsageConsumptionReport fields

bucket	A BucketRefOrValue. Bucket(s) included in the offer or option subscribed.
creationDate	A DateTime. Date and time of the request creation.
description	A String. Free short text describing the usage consumption content.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
lastUpdate	A DateTime. Date when the status was last changed.
logicalResource	A LogicalResourceRef. Reference to the LogicalResource eg MSISDN.
name	A String. Usage consumption name.
partyAccount	A PartyAccountRef. A reference to the account that owns the bucket.
product	A UsageConsumptionProductRef.
relatedParty	A RelatedPartyRefOrPartyRoleRef. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.
validPeriod	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Bucket sub-resource fields

See [Bucket] for details.

BucketCounter sub-resource fields

See [BucketCounter] for details.

Characteristic sub-resource fields

See [Characteristic] for details.

CharacteristicRelationship sub-resource fields

See [CharacteristicRelationship] for details.

BooleanArrayCharacteristic sub-resource fields

See [BooleanArrayCharacteristic] for details.

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BooleanCharacteristic sub-resource fields

See [BooleanCharacteristic] for details.

FloatArrayCharacteristic sub-resource fields

See [FloatArrayCharacteristic] for details.

FloatCharacteristic sub-resource fields

See [FloatCharacteristic] for details.

IntegerArrayCharacteristic sub-resource fields

See [IntegerArrayCharacteristic] for details.

IntegerCharacteristic sub-resource fields

See [IntegerCharacteristic] for details.

NumberArrayCharacteristic sub-resource fields

See [NumberArrayCharacteristic] for details.

NumberCharacteristic sub-resource fields

See [NumberCharacteristic] for details.

ObjectArrayCharacteristic sub-resource fields

See [ObjectArrayCharacteristic] for details.

ObjectCharacteristic sub-resource fields

See [ObjectCharacteristic] for details.

StringArrayCharacteristic sub-resource fields

See [StringArrayCharacteristic] for details.

StringCharacteristic sub-resource fields

See [StringCharacteristic] for details.

RelatedPartyRefOrPartyRoleRef sub-resource fields

See [RelatedPartyRefOrPartyRoleRef] for details.

PartyRef sub-resource fields

See [PartyRef] for details.

PartyRoleRef sub-resource fields

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See [PartyRoleRef] for details.

ConsumptionSummary sub-resource fields

See [ConsumptionSummary] for details.

BucketRelationship sub-resource fields

See [BucketRelationship] for details.

EntityRef sub-resource fields

See [EntityRef] for details.

EntityRelationship sub-resource fields

See [EntityRelationship] for details.

BucketSpecificationRef sub-resource fields

See [BucketSpecificationRef] for details.

LogicalResourceRef sub-resource fields

See [LogicalResourceRef] for details.

PartyAccountRef sub-resource fields

See [PartyAccountRef] for details.

ProductRef sub-resource fields

See [ProductRef] for details.

UsageConsumptionProductRef sub-resource fields

See [UsageConsumptionProductRef] for details.

RelatedPartyRefOrPartyRoleRef sub-resource fields

See [RelatedPartyRefOrPartyRoleRef] for details.

ReserveBalanceRef sub-resource fields

See [ReserveBalanceRef] for details.

BalanceActionRef sub-resource fields

See [BalanceActionRef] for details.

BucketRef sub-resource fields

See [BucketRef] for details.

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LogicalResourceRef sub-resource fields

See [LogicalResourceRef] for details.

PartyAccountRef sub-resource fields

See [PartyAccountRef] for details.

UsageConsumptionProductRef sub-resource fields

See [UsageConsumptionProductRef] for details.

ProductRef sub-resource fields

See [ProductRef] for details.

RelatedPartyRefOrPartyRoleRef sub-resource fields

See [RelatedPartyRefOrPartyRoleRef] for details.

Json representation sample(s)

The following example demonstrates the details of the UsageConsumptionReport resource.


```
{
  "id": "ucr0001",
  "href": "https://serverRoot/tmf-
api/usageConsumption/v4/usageConsumptionReport/ucr0001",
  "name": "Usage consumption report ucr0001",
  "description": "Usage consumption Report for Kate smartphone",
  "effectiveDate": "2018-03-15T15:44:28Z",
  "product": [
    {
      "id": "product1",
      "href": "https://serverRoot/tmf-api/productInventory/v5/product/product1",
      "name": "Kate smartphone",
      "consumptionSummary": [
        {
          "counterType": "outOfBucket",
          "level": "global",
          "value": {
            "amount": 20,
            "units": "USD"
          },
          "valueName": "20 USD",
          "consumptionPeriod": {
            "startDateTime": "2018-03-01T00:00:00Z",
            "endDateTime": "2018-03-15T00:00:00Z"
          },
          "user": {
            "role": "user",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
              "id": "usr1",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/usr1",
              "name": "Kate",
              "@type": "PartyRef",
```


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```
              "@referredType": "Individual"
            }
          },
          "@type": "ConsumptionSummary"
        }
      ],
      "@type": "ProductRef"
    }
  ],
  "bucket": [
    {
      "id": "bkt001",
      "href": "https://serverRoot/tmf-api/balanceManagement/v4/bucket/ucr0001",
      "name": "Main offer - data",
      "usageType": "data",
      "isShared": false,
      "product": [
        {
          "id": "product1",
          "href": "https://serverRoot/tmf-api/productInventory/v5/product/product1",
          "name": "Kate smartphone",
          "@type": "ProductRef"
        }
      ],
      "remainingValue": {
        "amount": 1.8,
        "units": "Go"
      },
      "remainingValueName": "it remains 1.8 Go",
      "validFor": {
        "startDateTime": "2018-03-15T00:00:00Z",
        "endDateTime": "2018-03-30T00:00:00Z"
      },
      "bucketCounter": [
        {
          "counterType": "used",
          "level": "global",
          "value": {
            "amount": 1.2,
            "units": "Go"
          },
          "valueName": "1.2 Go used",
          "consumptionPeriod": {
            "startDateTime": "2018-03-01T00:00:00Z",
            "endDateTime": "2018-03-15T00:00:00Z"
          },
          "user": {
            "role": "user",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
              "id": "usr1",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/usr1",
              "name": "Kate",
              "@type": "PartyRef",
              "@referredType": "Individual"
            }
          },
          "@type": "BucketCounter"
        }
      ],
      "@type": "Bucket"
    },
```


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```
    {
      "id": "bkt002",
      "href": "https://serverRoot/tmf-api/balanceManagement/v4/bucket/ucr0001",
      "name": "Main offer - national voice",
      "usageType": "national voice",
      "isShared": false,
      "product": [
        {
          "id": "product1",
          "href": "https://serverRoot/tmf-api/productInventory/v5/product/product1",
          "name": "Kate smartphone",
          "@type": "ProductRef"
        }
      ],
      "remainingValue": {
        "amount": 80,
        "units": "mins"
      },
      "remainingValueName": "it remains 80 minutes",
      "validFor": {
        "startDateTime": "2018-03-15T00:00:00Z",
        "endDateTime": "2018-03-30T00:00:00Z"
      },
      "bucketCounter": [
        {
          "counterType": "used",
          "level": "global",
          "value": {
            "amount": 40,
            "units": "mins"
          },
          "valueName": "40 mins used",
          "consumptionPeriod": {
            "startDateTime": "2018-03-01T00:00:00Z",
            "endDateTime": "2018-03-15T00:00:00Z"
          },
          "user": {
            "role": "user",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
              "id": "usr1",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/usr1",
              "name": "Kate",
              "@type": "PartyRef",
              "@referredType": "Individual"
            }
          },
          "@type": "BucketCounter"
        }
      ],
      "@type": "Bucket"
    },
    {
      "id": "bkt003",
      "href": "https://serverRoot/tmf-api/balanceManagement/v4/bucket/ucr0001",
      "name": "Main offer - sms",
      "usageType": "sms",
      "isShared": false,
      "product": [
        {
          "id": "product1",
          "href": "https://serverRoot/tmf-api/productInventory/v5/product/product1",
```


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```
            "name": "Kate smartphone",
            "@type": "ProductRef"
          }
        ],
        "remainingValue": {
          "amount": 95,
          "units": "sms"
        },
        "remainingValueName": "it remains 95 sms",
        "validFor": {
          "startDateTime": "2018-03-15T00:00:00Z",
          "endDateTime": "2018-03-30T00:00:00Z"
        },
        "bucketCounter": [
          {
            "counterType": "used",
            "level": "global",
            "value": {
              "amount": 25,
              "units": "sms"
            },
            "valueName": "25 sms used",
            "consumptionPeriod": {
              "startDateTime": "2018-03-01T00:00:00Z",
              "endDateTime": "2018-03-15T00:00:00Z"
            },
            "user": {
              "role": "user",
              "@type": "RelatedPartyRefOrPartyRoleRef",
              "partyOrPartyRole": {
                "id": "usr1",
                "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/usr1",
                "name": "Kate",
                "@type": "PartyRef",
                "@referredType": "Individual"
              }
            },
            "@type": "BucketCounter"
          }
        ],
        "@type": "Bucket"
      },
      {
        "id": "bkt004",
        "href": "https://serverRoot/tmf-api/balanceManagement/v4/bucket/ucr0001",
        "name": "Option Canada/USA - voice",
        "usageType": "voice",
        "isShared": false,
        "product": [
          {
            "id": "product1",
            "href": "https://serverRoot/tmf-api/productInventory/v5/product/product1",
            "name": "Kate smartphone",
            "@type": "ProductRef"
          }
        ],
        "remainingValue": {
          "amount": 10,
          "units": "mins"
        },
        "remainingValueName": "it remains 10 minutes",
        "validFor": {
```


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```
        "startDateTime": "2018-03-15T00:00:00Z",
        "endDateTime": "2018-03-30T00:00:00Z"
      },
      "bucketCounter": [
        {
          "counterType": "used",
          "level": "global",
          "value": {
            "amount": 20,
            "units": "mins"
          },
          "valueName": "20 mins used",
          "consumptionPeriod": {
            "startDateTime": "2018-03-01T00:00:00Z",
            "endDateTime": "2018-03-15T00:00:00Z"
          },
          "user": {
            "role": "user",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
              "id": "usr1",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/usr1",
              "name": "Kate",
              "@type": "PartyRef",
              "@referredType": "Individual"
            }
          },
          "@type": "BucketCounter"
        }
      ],
      "@type": "Bucket"
    },
    {
      "id": "bkt005",
      "href": "https://serverRoot/tmf-api/balanceManagement/v4/bucket/ucr0001",
      "name": "Option Canada/USA - sms",
      "usageType": "sms",
      "isShared": false,
      "product": [
        {
          "id": "product1",
          "href": "https://serverRoot/tmf-api/productInventory/v5/product/product1",
          "name": "Kate smartphone",
          "@type": "ProductRef"
        }
      ],
      "remainingValue": {
        "amount": 0,
        "units": "sms"
      },
      "remainingValueName": "Sms bucket exhausted",
      "validFor": {
        "startDateTime": "2018-03-15T00:00:00Z",
        "endDateTime": "2018-03-30T00:00:00Z"
      },
      "bucketCounter": [
        {
          "counterType": "used",
          "level": "global",
          "value": {
            "amount": 10,
            "units": "sms"
```


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```
          },
          "valueName": "10 sms used",
          "consumptionPeriod": {
            "startDateTime": "2018-03-01T00:00:00Z",
            "endDateTime": "2018-03-15T00:00:00Z"
          },
          "user": {
            "role": "user",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
              "id": "usr1",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/usr1",
              "name": "Kate",
              "@type": "PartyRef",
              "@referredType": "Individual"
            }
          },
          "@type": "BucketCounter"
        }
      ],
      "@type": "Bucket"
    }
  ],
  "@type": "UsageConsumptionReport"
}
```


Notification Resource Models

5 notifications are defined for this API.

Notifications related to QueryUsageConsumption:

- Create Event
- Delete Event
- State Change Event

Notifications related to UsageConsumptionReport:

- Create Event
- Delete Event

The notification structure for all notifications in this API follow the pattern depicted by the figure below. A notification event resource (depicted by "SpecificEvent" placeholder) is a sub class of a generic Event structure containing at least an id of the event occurrence (eventId), an event timestamp (eventTime), and the name of the resource (eventType). This notification structure owns an event payload structure ("SpecificEventPayload" placeholder) linked to the resource concerned by the notification using the resource name as access field ("resourceName" placeholder).

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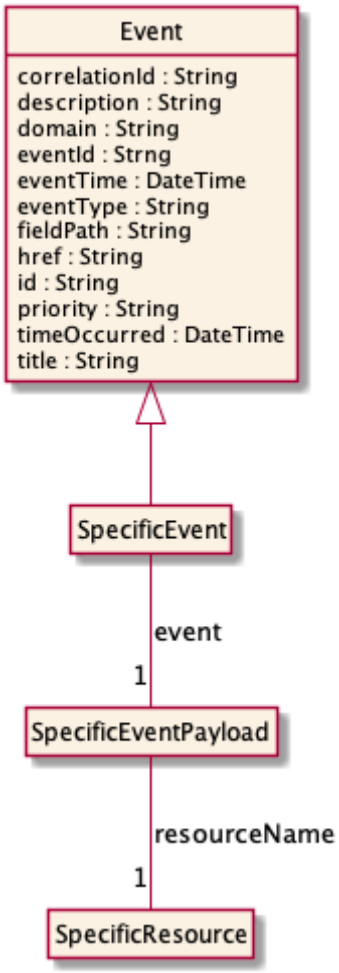


Figure 6 Notification Pattern

QueryUsageConsumption

Create Event

Message example for QueryUsageConsumptionCreateEvent event

```
Content-Type: application/json
{
  "correlationId": "3cb1acc3-e7cc",
  "description": "QueryUsageConsumption CreateEvent illustration",
  "domain": "Commercial",
  "eventId": "410d-b31d-62329cd302e1",
  "eventTime": "2022-08-25T12:25:47.711Z",
  "eventType": "QueryUsageConsumptionCreateEvent",
  "priority": "4",
  "timeOccurred": "2022-08-25T12:25:45.717Z",
  "title": "QueryUsageConsumptionCreateEvent",
  "event": {
    "queryUsageConsumption": {
      "id": "7655",
      "href": "https://mycsp.com:8080/tmf-
api/usageConsumption/v5/queryUsageConsumption/7655",
      "state": "done",
      "relatedParty": [
```

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```
        {
          "role": "Agent",
          "@type": "RelatedPartyRefOrPartyRoleRef",
          "partyOrPartyRole": {
            "id": "3426",
            "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/3426",
            "name": "Bob Smith",
            "@type": "PartyRef",
            "@referredType": "Individual"
          }
        }
      ],
      "searchCriteria": {
        "bucket": [
          {
            "usageType": "voice national",
            "@type": "Bucket"
          },
          {
            "usageType": "voice international",
            "@type": "Bucket"
          },
          {
            "usageType": "data",
            "@type": "Bucket"
          }
        ],
        "@type": "UsageConsumptionReport"
      },
      "usageConsumption": [
        {
          "id": "UC1",
          "creationDate": "2021-06-18T16:04:09.084Z",
          "lastUpdate": "2021-06-18T16:04:09.084Z",
          "bucket": [
            {
              "id": "BK1",
              "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/bucket/BK1",
              "description": "Bucket related to international calls",
              "isShared": false,
              "name": "RED Voice Plus Bucket - International calls",
              "remainingValue": {
                "amount": 2000,
                "units": "minutes"
              },
              "remainingValueName": "2000 minutes",
              "usageType": "voice international",
              "bucketCounter": [
                {
                  "id": "CT1",
                  "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT1",
                  "counterType": "used",
                  "level": "global",
                  "consumptionPeriod": {
                    "endDateTime": "2021-06-01T16:04:09.084Z",
                    "startDateTime": "2021-06-18T16:04:09.084Z"
                  },
                  "value": {
                    "amount": 1000,
                    "units": "minutes"
```


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```
                  },
                  "@type": "BucketCounter"
                }
              ],
              "bucketRelationship": [
                {
                  "relationshipType": "aggregated",
                  "bucket": {
                    "id": "BK2",
                    "href": "https://mycsp.com:8080/tmf-
api/usageConsumption/v5/bucket/BK2",
                    "@type": "BucketRef"
                  },
                  "@type": "BucketRelationship"
                }
              ],
              "product": [
                {
                  "id": "PR1",
                  "href": "https://mycsp.com:8080/tmf-
api/productInventory/v5/product/PR1",
                  "name": "RED Voice Plus",
                  "@type": "ProductRef"
                }
              ],
              "relatedParty": [
                {
                  "role": "Customer",
                  "@type": "RelatedPartyRefOrPartyRoleRef",
                  "partyOrPartyRole": {
                    "id": "3426",
                    "href": "https://mycsp.com:8080/tmf-
api/customerManagement/v5/customer/3426",
                    "name": "John Doe",
                    "@type": "PartyRoleRef",
                    "@referredType": "Customer"
                  }
                }
              ],
              "@type": "Bucket"
            },
            {
              "id": "BK2",
              "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/bucket/BK2",
              "description": "Bucket related to national calls",
              "isShared": false,
              "name": "RED Voice Plus Bucket - National calls",
              "remainingValue": {
                "amount": 4000,
                "units": "minutes"
              },
              "remainingValueName": "4000 minutes",
              "usageType": "voice national",
              "bucketCounter": [
                {
                  "id": "CT2",
                  "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT2",
                  "counterType": "used",
                  "level": "global",
                  "consumptionPeriod": {
                    "endDateTime": "2021-06-01T16:04:09.084Z",
```


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```
                    "startDateTime": "2021-06-18T16:04:09.084Z"
                  },
                  "value": {
                    "amount": 1000,
                    "units": "minutes"
                  },
                  "@type": "ConsumptionSummary"
                }
              ],
              "bucketRelationship": [
                {
                  "relationshipType": "aggregated",
                  "bucket": {
                    "id": "BK1",
                    "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/bucket/BK2",
                    "@type": "BucketRef"
                  },
                  "@type": "BucketRelationship"
                }
              ],
              "product": [
                {
                  "id": "PR1",
                  "href": "https://mycsp.com:8080/tmf-
api/productInventory/v5/product/PR1",
                  "name": "RED Voice Plus",
                  "@type": "ProductRef"
                }
              ],
              "relatedParty": [
                {
                  "role": "Customer",
                  "@type": "RelatedPartyRefOrPartyRoleRef",
                  "partyOrPartyRole": {
                    "id": "3426",
                    "href": "https://mycsp.com:8080/tmf-
api/customerManagement/v5/customer/3426",
                    "name": "John Doe",
                    "@type": "PartyRoleRef",
                    "@referredType": "Customer"
                  }
                }
              ],
              "@type": "Bucket"
            },
            {
              "id": "BK3",
              "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/bucket/BK3",
              "description": "Bucket related to data",
              "isShared": true,
              "name": "RED Shared Data 10 - Data",
              "usageType": "data",
              "bucketCounter": [
                {
                  "id": "CT3",
                  "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT3",
                  "counterType": "used",
                  "level": "detailed by user",
                  "consumptionPeriod": {
                    "endDateTime": "2021-06-01T16:04:09.084Z",
```


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```
                    "startDateTime": "2021-06-18T16:04:09.084Z"
                  },
                  "user": {
                    "role": "Shared data member",
                    "@type": "RelatedPartyRefOrPartyRoleRef",
                    "partyOrPartyRole": {
                      "id": "CUST2",
                      "name": "Helen Water",
                      "@type": "PartyRef",
                      "@referredType": "Individual"
                    }
                  },
                  "value": {
                    "amount": 1,
                    "units": "GB"
                  },
                  "@type": "ConsumptionSummary"
                },
                {
                  "id": "CT4",
                  "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT4",
                  "counterType": "used",
                  "level": "detailed by user",
                  "consumptionPeriod": {
                    "endDateTime": "2021-06-01T16:04:09.084Z",
                    "startDateTime": "2021-06-18T16:04:09.084Z"
                  },
                  "user": {
                    "role": "Shared data member",
                    "@type": "RelatedPartyRefOrPartyRoleRef",
                    "partyOrPartyRole": {
                      "id": "CUST3",
                      "name": "Shawn Water",
                      "@type": "PartyRef",
                      "@referredType": "Individual"
                    }
                  },
                  "value": {
                    "amount": 0.7,
                    "units": "GB"
                  },
                  "@type": "ConsumptionSummary"
                },
                {
                  "id": "CT5",
                  "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT5",
                  "counterType": "used",
                  "level": "detailed by user",
                  "consumptionPeriod": {
                    "endDateTime": "2021-06-01T16:04:09.084Z",
                    "startDateTime": "2021-06-18T16:04:09.084Z"
                  },
                  "user": {
                    "role": "Shared data member",
                    "@type": "RelatedPartyRefOrPartyRoleRef",
                    "partyOrPartyRole": {
                      "id": "CUST1",
                      "name": "John Water",
                      "@type": "PartyRef",
                      "@referredType": "Individual"
                    }
```


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```
                    },
                    "value": {
                      "amount": 1.5,
                      "units": "GB"
                    },
                    "@type": "ConsumptionSummary"
                  },
                  {
                    "id": "CT6",
                    "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT6",
                    "counterType": "remaining",
                    "level": "detailed by user",
                    "consumptionPeriod": {
                      "endDateTime": "2021-06-01T16:04:09.084Z",
                      "startDateTime": "2021-06-18T16:04:09.084Z"
                    },
                    "user": {
                      "role": "Shared data member",
                      "@type": "RelatedPartyRefOrPartyRoleRef",
                      "partyOrPartyRole": {
                        "id": "CUST2",
                        "name": "Helen Water",
                        "@type": "PartyRef",
                        "@referredType": "Individual"
                      }
                    },
                    "value": {
                      "amount": 1,
                      "units": "GB"
                    },
                    "@type": "ConsumptionSummary"
                  },
                  {
                    "id": "CT7",
                    "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT7",
                    "counterType": "remaining",
                    "level": "detailed by user",
                    "consumptionPeriod": {
                      "endDateTime": "2021-06-01T16:04:09.084Z",
                      "startDateTime": "2021-06-18T16:04:09.084Z"
                    },
                    "user": {
                      "role": "Shared data member",
                      "@type": "RelatedPartyRefOrPartyRoleRef",
                      "partyOrPartyRole": {
                        "id": "CUST3",
                        "name": "Shawn Water",
                        "@type": "PartyRef",
                        "@referredType": "Individual"
                      }
                    },
                    "value": {
                      "amount": 1.3,
                      "units": "GB"
                    },
                    "@type": "ConsumptionSummary"
                  },
                  {
                    "id": "CT8",
                    "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT8",
```


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```
                  "counterType": "remaining",
                  "level": "detailed by user",
                  "consumptionPeriod": {
                    "endDateTime": "2021-06-01T16:04:09.084Z",
                    "startDateTime": "2021-06-18T16:04:09.084Z"
                  },
                  "user": {
                    "role": "Shared data member",
                    "@type": "RelatedPartyRefOrPartyRoleRef",
                    "partyOrPartyRole": {
                      "id": "CUST1",
                      "name": "John Water",
                      "@type": "PartyRef",
                      "@referredType": "Individual"
                    }
                  },
                  "value": {
                    "amount": 4.5,
                    "units": "GB"
                  },
                  "@type": "ConsumptionSummary"
                }
              ],
              "product": [
                {
                  "id": "PR2",
                  "href": "https://mycsp.com:8080/tmf-
api/productInventory/v5/product/PR2",
                  "name": "RED Shared Data 10",
                  "@type": "ProductRef"
                }
              ],
              "relatedParty": [
                {
                  "role": "Customer",
                  "@type": "RelatedPartyRefOrPartyRoleRef",
                  "partyOrPartyRole": {
                    "id": "3426",
                    "href": "https://mycsp.com:8080/tmf-
api/customerManagement/v5/customer/3426",
                    "name": "John Doe",
                    "@type": "PartyRoleRef",
                    "@referredType": "Customer"
                  }
                }
              ],
              "@type": "Bucket"
            }
          ],
          "@type": "UsageConsumptionReport"
        }
      ],
      "@type": "QueryUsageConsumption"
    }
  },
  "reportingSystem": {
    "id": "694",
    "name": "APP-764",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "621",
```


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```
    "name": "APP-625",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "QueryUsageConsumptionCreateEvent"
}
```

Delete Event

Message example for QueryUsageConsumptionDeleteEvent event

```
Content-Type: application/json
{
  "correlationId": "3cb1acc3-e7fe",
  "description": "QueryUsageConsumption DeleteEvent illustration",
  "domain": "Commercial",
  "eventId": "410d-b31d-62329cd302e8",
  "eventTime": "2022-08-25T12:25:47.711Z",
  "eventType": "QueryUsageConsumptionDeleteEvent",
  "priority": "4",
  "timeOccurred": "2022-08-25T12:25:45.717Z",
  "title": "QueryUsageConsumptionDeleteEvent",
  "event": {
    "queryUsageConsumption": {
      "id": "7655",
      "href": "https://mycsp.com:8080/tmf-
api/usageConsumption/v5/queryUsageConsumption/7655",
      "state": "done",
      "relatedParty": [
        {
          "role": "Agent",
          "@type": "RelatedPartyRefOrPartyRoleRef",
          "partyOrPartyRole": {
            "id": "3426",
            "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/3426",
            "name": "Bob Smith",
            "@type": "PartyRef",
            "@referredType": "Individual"
          }
        }
      ],
      "searchCriteria": {
        "bucket": [
          {
            "usageType": "voice national",
            "@type": "Bucket"
          },
          {
            "usageType": "voice international",
            "@type": "Bucket"
          },
          {
            "usageType": "data",
            "@type": "Bucket"
          }
        ],
        "@type": "UsageConsumptionReport"
      },
```

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```
      "usageConsumption": [
        {
          "id": "UC1",
          "creationDate": "2021-06-18T16:04:09.084Z",
          "lastUpdate": "2021-06-18T16:04:09.084Z",
          "bucket": [
            {
              "id": "BK1",
              "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/bucket/BK1",
              "description": "Bucket related to international calls",
              "isShared": false,
              "name": "RED Voice Plus Bucket - International calls",
              "remainingValue": {
                "amount": 2000,
                "units": "minutes"
              },
              "remainingValueName": "2000 minutes",
              "usageType": "voice international",
              "bucketCounter": [
                {
                  "id": "CT1",
                  "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT1",
                  "counterType": "used",
                  "level": "global",
                  "consumptionPeriod": {
                    "endDateTime": "2021-06-01T16:04:09.084Z",
                    "startDateTime": "2021-06-18T16:04:09.084Z"
                  },
                  "value": {
                    "amount": 1000,
                    "units": "minutes"
                  },
                  "@type": "BucketCounter"
                }
              ],
              "bucketRelationship": [
                {
                  "relationshipType": "aggregated",
                  "bucket": {
                    "id": "BK2",
                    "href": "https://mycsp.com:8080/tmf-
api/usageConsumption/v5/bucket/BK2",
                    "@type": "BucketRef"
                  },
                  "@type": "BucketRelationship"
                }
              ],
              "product": [
                {
                  "id": "PR1",
                  "href": "https://mycsp.com:8080/tmf-
api/productInventory/v5/product/PR1",
                  "name": "RED Voice Plus",
                  "@type": "ProductRef"
                }
              ],
              "relatedParty": [
                {
                  "role": "Customer",
                  "@type": "RelatedPartyRefOrPartyRoleRef",
                  "partyOrPartyRole": {
```


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```
                    "id": "3426",
                    "href": "https://mycsp.com:8080/tmf-
api/customerManagement/v5/customer/3426",
                    "name": "John Doe",
                    "@type": "PartyRoleRef",
                    "@referredType": "Customer"
                  }
                }
              ],
              "@type": "Bucket"
            },
            {
              "id": "BK2",
              "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/bucket/BK2",
              "description": "Bucket related to national calls",
              "isShared": false,
              "name": "RED Voice Plus Bucket - National calls",
              "remainingValue": {
                "amount": 4000,
                "units": "minutes"
              },
              "remainingValueName": "4000 minutes",
              "usageType": "voice national",
              "bucketCounter": [
                {
                  "id": "CT2",
                  "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT2",
                  "counterType": "used",
                  "level": "global",
                  "consumptionPeriod": {
                    "endDateTime": "2021-06-01T16:04:09.084Z",
                    "startDateTime": "2021-06-18T16:04:09.084Z"
                  },
                  "value": {
                    "amount": 1000,
                    "units": "minutes"
                  },
                  "@type": "ConsumptionSummary"
                }
              ],
              "bucketRelationship": [
                {
                  "relationshipType": "aggregated",
                  "bucket": {
                    "id": "BK1",
                    "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/bucket/BK2",
                    "@type": "BucketRef"
                  },
                  "@type": "BucketRelationship"
                }
              ],
              "product": [
                {
                  "id": "PR1",
                  "href": "https://mycsp.com:8080/tmf-
api/productInventory/v5/product/PR1",
                  "name": "RED Voice Plus",
                  "@type": "ProductRef"
                }
              ],
```


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```
            "relatedParty": [
              {
                "role": "Customer",
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "partyOrPartyRole": {
                  "id": "3426",
                  "href": "https://mycsp.com:8080/tmf-
api/customerManagement/v5/customer/3426",
                  "name": "John Doe",
                  "@type": "PartyRoleRef",
                  "@referredType": "Customer"
                }
              }
            ],
            "@type": "Bucket"
          },
          {
            "id": "BK3",
            "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/bucket/BK3",
            "description": "Bucket related to data",
            "isShared": true,
            "name": "RED Shared Data 10 - Data",
            "usageType": "data",
            "bucketCounter": [
              {
                "id": "CT3",
                "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT3",
                "counterType": "used",
                "level": "detailed by user",
                "consumptionPeriod": {
                  "endDateTime": "2021-06-01T16:04:09.084Z",
                  "startDateTime": "2021-06-18T16:04:09.084Z"
                },
                "user": {
                  "role": "Shared data member",
                  "@type": "RelatedPartyRefOrPartyRoleRef",
                  "partyOrPartyRole": {
                    "id": "CUST2",
                    "name": "Helen Water",
                    "@type": "PartyRef",
                    "@referredType": "Individual"
                  }
                },
                "value": {
                  "amount": 1,
                  "units": "GB"
                },
                "@type": "ConsumptionSummary"
              },
              {
                "id": "CT4",
                "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT4",
                "counterType": "used",
                "level": "detailed by user",
                "consumptionPeriod": {
                  "endDateTime": "2021-06-01T16:04:09.084Z",
                  "startDateTime": "2021-06-18T16:04:09.084Z"
                },
                "user": {
                  "role": "Shared data member",
```


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```
                    "@type": "RelatedPartyRefOrPartyRoleRef",
                    "partyOrPartyRole": {
                      "id": "CUST3",
                      "name": "Shawn Water",
                      "@type": "PartyRef",
                      "@referredType": "Individual"
                    }
                  },
                  "value": {
                    "amount": 0.7,
                    "units": "GB"
                  },
                  "@type": "ConsumptionSummary"
                },
                {
                  "id": "CT5",
                  "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT5",
                  "counterType": "used",
                  "level": "detailed by user",
                  "consumptionPeriod": {
                    "endDateTime": "2021-06-01T16:04:09.084Z",
                    "startDateTime": "2021-06-18T16:04:09.084Z"
                  },
                  "user": {
                    "role": "Shared data member",
                    "@type": "RelatedPartyRefOrPartyRoleRef",
                    "partyOrPartyRole": {
                      "id": "CUST1",
                      "name": "John Water",
                      "@type": "PartyRef",
                      "@referredType": "Individual"
                    }
                  },
                  "value": {
                    "amount": 1.5,
                    "units": "GB"
                  },
                  "@type": "ConsumptionSummary"
                },
                {
                  "id": "CT6",
                  "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT6",
                  "counterType": "remaining",
                  "level": "detailed by user",
                  "consumptionPeriod": {
                    "endDateTime": "2021-06-01T16:04:09.084Z",
                    "startDateTime": "2021-06-18T16:04:09.084Z"
                  },
                  "user": {
                    "role": "Shared data member",
                    "@type": "RelatedPartyRefOrPartyRoleRef",
                    "partyOrPartyRole": {
                      "id": "CUST2",
                      "name": "Helen Water",
                      "@type": "PartyRef",
                      "@referredType": "Individual"
                    }
                  },
                  "value": {
                    "amount": 1,
                    "units": "GB"
```


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```
                  },
                  "@type": "ConsumptionSummary"
                },
                {
                  "id": "CT7",
                  "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT7",
                  "counterType": "remaining",
                  "level": "detailed by user",
                  "consumptionPeriod": {
                    "endDateTime": "2021-06-01T16:04:09.084Z",
                    "startDateTime": "2021-06-18T16:04:09.084Z"
                  },
                  "user": {
                    "role": "Shared data member",
                    "@type": "RelatedPartyRefOrPartyRoleRef",
                    "partyOrPartyRole": {
                      "id": "CUST3",
                      "name": "Shawn Water",
                      "@type": "PartyRef",
                      "@referredType": "Individual"
                    }
                  },
                  "value": {
                    "amount": 1.3,
                    "units": "GB"
                  },
                  "@type": "ConsumptionSummary"
                },
                {
                  "id": "CT8",
                  "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT8",
                  "counterType": "remaining",
                  "level": "detailed by user",
                  "consumptionPeriod": {
                    "endDateTime": "2021-06-01T16:04:09.084Z",
                    "startDateTime": "2021-06-18T16:04:09.084Z"
                  },
                  "user": {
                    "role": "Shared data member",
                    "@type": "RelatedPartyRefOrPartyRoleRef",
                    "partyOrPartyRole": {
                      "id": "CUST1",
                      "name": "John Water",
                      "@type": "PartyRef",
                      "@referredType": "Individual"
                    }
                  },
                  "value": {
                    "amount": 4.5,
                    "units": "GB"
                  },
                  "@type": "ConsumptionSummary"
                }
              ],
              "product": [
                {
                  "id": "PR2",
                  "href": "https://mycsp.com:8080/tmf-
api/productInventory/v5/product/PR2",
                  "name": "RED Shared Data 10",
                  "@type": "ProductRef"
```


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```
                }
              ],
              "relatedParty": [
                {
                  "role": "Customer",
                  "@type": "RelatedPartyRefOrPartyRoleRef",
                  "partyOrPartyRole": {
                    "id": "3426",
                    "href": "https://mycsp.com:8080/tmf-
api/customerManagement/v5/customer/3426",
                    "name": "John Doe",
                    "@type": "PartyRoleRef",
                    "@referredType": "Customer"
                  }
                }
              ],
              "@type": "Bucket"
            }
          ],
          "@type": "UsageConsumptionReport"
        }
      ],
      "@type": "QueryUsageConsumption"
    }
  },
  "reportingSystem": {
    "id": "694",
    "name": "APP-764",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "621",
    "name": "APP-625",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "QueryUsageConsumptionDeleteEvent"
}
```


State Change Event

Message example for QueryUsageConsumptionStateChangeEvent event


```
Content-Type: application/json
{
  "correlationId": "3cb1acc3-e7cc",
  "description": "QueryUsageConsumption StateChangeEvent illustration",
  "domain": "Commercial",
  "eventId": "410d-b31d-62329cd302e1",
  "eventTime": "2022-08-25T12:25:47.711Z",
  "eventType": "QueryUsageConsumptionStateChangeEvent",
  "priority": "4",
  "timeOccurred": "2022-08-25T12:25:45.717Z",
  "title": "QueryUsageConsumptionStateChangeEvent",
  "event": {
    "queryUsageConsumption": {
      "id": "7655",
      "href": "https://mycsp.com:8080/tmf-
```


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```
api/usageConsumption/v5/queryUsageConsumption/7655",
      "state": "done",
      "relatedParty": [
        {
          "role": "Agent",
          "@type": "RelatedPartyRefOrPartyRoleRef",
          "partyOrPartyRole": {
            "id": "3426",
            "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/3426",
            "name": "Bob Smith",
            "@type": "PartyRef",
            "@referredType": "Individual"
          }
        }
      ],
      "searchCriteria": {
        "bucket": [
          {
            "usageType": "voice national",
            "@type": "Bucket"
          },
          {
            "usageType": "voice international",
            "@type": "Bucket"
          },
          {
            "usageType": "data",
            "@type": "Bucket"
          }
        ],
        "@type": "UsageConsumptionReport"
      },
      "usageConsumption": [
        {
          "id": "UC1",
          "creationDate": "2021-06-18T16:04:09.084Z",
          "lastUpdate": "2021-06-18T16:04:09.084Z",
          "bucket": [
            {
              "id": "BK1",
              "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/bucket/BK1",
              "description": "Bucket related to international calls",
              "isShared": false,
              "name": "RED Voice Plus Bucket - International calls",
              "remainingValue": {
                "amount": 2000,
                "units": "minutes"
              },
              "remainingValueName": "2000 minutes",
              "usageType": "voice international",
              "bucketCounter": [
                {
                  "id": "CT1",
                  "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT1",
                  "counterType": "used",
                  "level": "global",
                  "consumptionPeriod": {
                    "endDateTime": "2021-06-01T16:04:09.084Z",
                    "startDateTime": "2021-06-18T16:04:09.084Z"
                  },
```


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```
                  "value": {
                    "amount": 1000,
                    "units": "minutes"
                  },
                  "@type": "BucketCounter"
                }
              ],
              "bucketRelationship": [
                {
                  "relationshipType": "aggregated",
                  "bucket": {
                    "id": "BK2",
                    "href": "https://mycsp.com:8080/tmf-
api/usageConsumption/v5/bucket/BK2",
                    "@type": "BucketRef"
                  },
                  "@type": "BucketRelationship"
                }
              ],
              "product": [
                {
                  "id": "PR1",
                  "href": "https://mycsp.com:8080/tmf-
api/productInventory/v5/product/PR1",
                  "name": "RED Voice Plus",
                  "@type": "ProductRef"
                }
              ],
              "relatedParty": [
                {
                  "role": "Customer",
                  "@type": "RelatedPartyRefOrPartyRoleRef",
                  "partyOrPartyRole": {
                    "id": "3426",
                    "href": "https://mycsp.com:8080/tmf-
api/customerManagement/v5/customer/3426",
                    "name": "John Doe",
                    "@type": "PartyRoleRef",
                    "@referredType": "Customer"
                  }
                }
              ],
              "@type": "Bucket"
            },
            {
              "id": "BK2",
              "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/bucket/BK2",
              "description": "Bucket related to national calls",
              "isShared": false,
              "name": "RED Voice Plus Bucket - National calls",
              "remainingValue": {
                "amount": 4000,
                "units": "minutes"
              },
              "remainingValueName": "4000 minutes",
              "usageType": "voice national",
              "bucketCounter": [
                {
                  "id": "CT2",
                  "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT2",
                  "counterType": "used",
```


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```
                    "level": "global",
                    "consumptionPeriod": {
                      "endDateTime": "2021-06-01T16:04:09.084Z",
                      "startDateTime": "2021-06-18T16:04:09.084Z"
                    },
                    "value": {
                      "amount": 1000,
                      "units": "minutes"
                    },
                    "@type": "ConsumptionSummary"
                  }
                ],
                "bucketRelationship": [
                  {
                    "relationshipType": "aggregated",
                    "bucket": {
                      "id": "BK1",
                      "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/bucket/BK2",
                      "@type": "BucketRef"
                    },
                    "@type": "BucketRelationship"
                  }
                ],
                "product": [
                  {
                    "id": "PR1",
                    "href": "https://mycsp.com:8080/tmf-
api/productInventory/v5/product/PR1",
                    "name": "RED Voice Plus",
                    "@type": "ProductRef"
                  }
                ],
                "relatedParty": [
                  {
                    "role": "Customer",
                    "@type": "RelatedPartyRefOrPartyRoleRef",
                    "partyOrPartyRole": {
                      "id": "3426",
                      "href": "https://mycsp.com:8080/tmf-
api/customerManagement/v5/customer/3426",
                      "name": "John Doe",
                      "@type": "PartyRoleRef",
                      "@referredType": "Customer"
                    }
                  }
                ],
                "@type": "Bucket"
              },
              {
                "id": "BK3",
                "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/bucket/BK3",
                "description": "Bucket related to data",
                "isShared": true,
                "name": "RED Shared Data 10 - Data",
                "usageType": "data",
                "bucketCounter": [
                  {
                    "id": "CT3",
                    "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT3",
                    "counterType": "used",
```


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```
                    "level": "detailed by user",
                    "consumptionPeriod": {
                      "endDateTime": "2021-06-01T16:04:09.084Z",
                      "startDateTime": "2021-06-18T16:04:09.084Z"
                    },
                    "user": {
                      "role": "Shared data member",
                      "@type": "RelatedPartyRefOrPartyRoleRef",
                      "partyOrPartyRole": {
                        "id": "CUST2",
                        "name": "Helen Water",
                        "@type": "PartyRef",
                        "@referredType": "Individual"
                      }
                    },
                    "value": {
                      "amount": 1,
                      "units": "GB"
                    },
                    "@type": "ConsumptionSummary"
                  },
                  {
                    "id": "CT4",
                    "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT4",
                    "counterType": "used",
                    "level": "detailed by user",
                    "consumptionPeriod": {
                      "endDateTime": "2021-06-01T16:04:09.084Z",
                      "startDateTime": "2021-06-18T16:04:09.084Z"
                    },
                    "user": {
                      "role": "Shared data member",
                      "@type": "RelatedPartyRefOrPartyRoleRef",
                      "partyOrPartyRole": {
                        "id": "CUST3",
                        "name": "Shawn Water",
                        "@type": "PartyRef",
                        "@referredType": "Individual"
                      }
                    },
                    "value": {
                      "amount": 0.7,
                      "units": "GB"
                    },
                    "@type": "ConsumptionSummary"
                  },
                  {
                    "id": "CT5",
                    "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT5",
                    "counterType": "used",
                    "level": "detailed by user",
                    "consumptionPeriod": {
                      "endDateTime": "2021-06-01T16:04:09.084Z",
                      "startDateTime": "2021-06-18T16:04:09.084Z"
                    },
                    "user": {
                      "role": "Shared data member",
                      "@type": "RelatedPartyRefOrPartyRoleRef",
                      "partyOrPartyRole": {
                        "id": "CUST1",
                        "name": "John Water",
```


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```
                    "@type": "PartyRef",
                    "@referredType": "Individual"
                  }
                },
                "value": {
                  "amount": 1.5,
                  "units": "GB"
                },
                "@type": "ConsumptionSummary"
              },
              {
                "id": "CT6",
                "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT6",
                "counterType": "remaining",
                "level": "detailed by user",
                "consumptionPeriod": {
                  "endDateTime": "2021-06-01T16:04:09.084Z",
                  "startDateTime": "2021-06-18T16:04:09.084Z"
                },
                "user": {
                  "role": "Shared data member",
                  "@type": "RelatedPartyRefOrPartyRoleRef",
                  "partyOrPartyRole": {
                    "id": "CUST2",
                    "name": "Helen Water",
                    "@type": "PartyRef",
                    "@referredType": "Individual"
                  }
                },
                "value": {
                  "amount": 1,
                  "units": "GB"
                },
                "@type": "ConsumptionSummary"
              },
              {
                "id": "CT7",
                "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT7",
                "counterType": "remaining",
                "level": "detailed by user",
                "consumptionPeriod": {
                  "endDateTime": "2021-06-01T16:04:09.084Z",
                  "startDateTime": "2021-06-18T16:04:09.084Z"
                },
                "user": {
                  "role": "Shared data member",
                  "@type": "RelatedPartyRefOrPartyRoleRef",
                  "partyOrPartyRole": {
                    "id": "CUST3",
                    "name": "Shawn Water",
                    "@type": "PartyRef",
                    "@referredType": "Individual"
                  }
                },
                "value": {
                  "amount": 1.3,
                  "units": "GB"
                },
                "@type": "ConsumptionSummary"
              },
              {
```


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```
                  "id": "CT8",
                  "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT8",
                  "counterType": "remaining",
                  "level": "detailed by user",
                  "consumptionPeriod": {
                    "endDateTime": "2021-06-01T16:04:09.084Z",
                    "startDateTime": "2021-06-18T16:04:09.084Z"
                  },
                  "user": {
                    "role": "Shared data member",
                    "@type": "RelatedPartyRefOrPartyRoleRef",
                    "partyOrPartyRole": {
                      "id": "CUST1",
                      "name": "John Water",
                      "@type": "PartyRef",
                      "@referredType": "Individual"
                    }
                  },
                  "value": {
                    "amount": 4.5,
                    "units": "GB"
                  },
                  "@type": "ConsumptionSummary"
                }
              ],
              "product": [
                {
                  "id": "PR2",
                  "href": "https://mycsp.com:8080/tmf-
api/productInventory/v5/product/PR2",
                  "name": "RED Shared Data 10",
                  "@type": "ProductRef"
                }
              ],
              "relatedParty": [
                {
                  "role": "Customer",
                  "@type": "RelatedPartyRefOrPartyRoleRef",
                  "partyOrPartyRole": {
                    "id": "3426",
                    "href": "https://mycsp.com:8080/tmf-
api/customerManagement/v5/customer/3426",
                    "name": "John Doe",
                    "@type": "PartyRoleRef",
                    "@referredType": "Customer"
                  }
                }
              ],
              "@type": "Bucket"
            }
          ],
          "@type": "UsageConsumptionReport"
        }
      ],
      "@type": "QueryUsageConsumption"
    }
  },
  "reportingSystem": {
    "id": "694",
    "name": "APP-764",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
```


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```
  },
  "source": {
    "id": "621",
    "name": "APP-625",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "QueryUsageConsumptionStateChangeEvent"
}
```

UsageConsumptionReport

Create Event

Message example for UsageConsumptionReportCreateEvent event

```
Content-Type: application/json
{
  "correlationId": "3cb1acc3-e7cc",
  "description": "UsageConsumptionReport CreateEvent illustration",
  "domain": "Commercial",
  "eventId": "410d-b31d-62329cd302e1",
  "eventTime": "2022-08-25T12:25:47.711Z",
  "eventType": "UsageConsumptionReportCreateEvent",
  "priority": "4",
  "timeOccurred": "2022-08-25T12:25:45.717Z",
  "title": "UsageConsumptionReportCreateEvent",
  "event": {
    "usageConsumptionReport": {
      "id": "UC1",
      "creationDate": "2021-06-18T16:04:09.084Z",
      "lastUpdate": "2021-06-18T16:04:09.084Z",
      "bucket": [
        {
          "id": "BK1",
          "href": "https://mycsp.com:8080/tmf-api/balanceManagement/v5/bucket/BK1",
          "description": "Bucket related to international calls",
          "isShared": false,
          "name": "RED Voice Plus Bucket - International calls",
          "remainingValue": {
            "amount": 2000,
            "units": "minutes"
          },
          "remainingValueName": "2000 minutes",
          "usageType": "voice international",
          "bucketCounter": [
            {
              "id": "CT1",
              "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT1",
              "counterType": "used",
              "level": "global",
              "consumptionPeriod": {
                "endDateTime": "2021-06-01T16:04:09.084Z",
                "startDateTime": "2021-06-18T16:04:09.084Z"
              },
              "value": {
                "amount": 1000,
```

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```
                "units": "minutes"
              },
              "@type": "BucketCounter"
            }
          ],
          "bucketRelationship": [
            {
              "relationshipType": "aggregated",
              "bucket": {
                "id": "BK2",
                "href": "https://mycsp.com:8080/tmf-
api/usageConsumption/v5/bucket/BK2",
                "@type": "BucketRef"
              },
              "@type": "BucketRelationship"
            }
          ],
          "product": [
            {
              "id": "PR1",
              "href": "https://mycsp.com:8080/tmf-
api/productInventory/v5/product/PR1",
              "name": "RED Voice Plus",
              "@type": "ProductRef"
            }
          ],
          "relatedParty": [
            {
              "role": "Customer",
              "@type": "RelatedPartyRefOrPartyRoleRef",
              "partyOrPartyRole": {
                "id": "3426",
                "href": "https://mycsp.com:8080/tmf-
api/customerManagement/v5/customer/3426",
                "name": "John Doe",
                "@type": "PartyRoleRef",
                "@referredType": "Customer"
              }
            }
          ],
          "@type": "Bucket"
        },
        {
          "id": "BK2",
          "href": "https://mycsp.com:8080/tmf-api/balanceManagement/v5/bucket/BK2",
          "description": "Bucket related to national calls",
          "isShared": false,
          "name": "RED Voice Plus Bucket - National calls",
          "remainingValue": {
            "amount": 4000,
            "units": "minutes"
          },
          "remainingValueName": "4000 minutes",
          "usageType": "voice national",
          "bucketCounter": [
            {
              "id": "CT2",
              "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT2",
              "counterType": "used",
              "level": "global",
              "consumptionPeriod": {
                "endDateTime": "2021-06-01T16:04:09.084Z",
```


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```
                "startDateTime": "2021-06-18T16:04:09.084Z"
              },
              "value": {
                "amount": 1000,
                "units": "minutes"
              },
              "@type": "ConsumptionSummary"
            }
          ],
          "bucketRelationship": [
            {
              "relationshipType": "aggregated",
              "bucket": {
                "id": "BK1",
                "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/bucket/BK2",
                "@type": "BucketRef"
              },
              "@type": "BucketRelationship"
            }
          ],
          "product": [
            {
              "id": "PR1",
              "href": "https://mycsp.com:8080/tmf-
api/productInventory/v5/product/PR1",
              "name": "RED Voice Plus",
              "@type": "ProductRef"
            }
          ],
          "relatedParty": [
            {
              "role": "Customer",
              "@type": "RelatedPartyRefOrPartyRoleRef",
              "partyOrPartyRole": {
                "id": "3426",
                "href": "https://mycsp.com:8080/tmf-
api/customerManagement/v5/customer/3426",
                "name": "John Doe",
                "@type": "PartyRoleRef",
                "@referredType": "Customer"
              }
            }
          ],
          "@type": "Bucket"
        },
        {
          "id": "BK3",
          "href": "https://mycsp.com:8080/tmf-api/balanceManagement/v5/bucket/BK3",
          "description": "Bucket related to data",
          "isShared": true,
          "name": "RED Shared Data 10 - Data",
          "usageType": "data",
          "bucketCounter": [
            {
              "id": "CT3",
              "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT3",
              "counterType": "used",
              "level": "detailed by user",
              "consumptionPeriod": {
                "endDateTime": "2021-06-01T16:04:09.084Z",
                "startDateTime": "2021-06-18T16:04:09.084Z"
```


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```
            },
            "user": {
              "role": "Shared data member",
              "@type": "RelatedPartyRefOrPartyRoleRef",
              "partyOrPartyRole": {
                "id": "CUST2",
                "name": "Helen Water",
                "@type": "PartyRef",
                "@referredType": "Individual"
              }
            },
            "value": {
              "amount": 1,
              "units": "GB"
            },
            "@type": "ConsumptionSummary"
          },
          {
            "id": "CT4",
            "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT4",
            "counterType": "used",
            "level": "detailed by user",
            "consumptionPeriod": {
              "endDateTime": "2021-06-01T16:04:09.084Z",
              "startDateTime": "2021-06-18T16:04:09.084Z"
            },
            "user": {
              "role": "Shared data member",
              "@type": "RelatedPartyRefOrPartyRoleRef",
              "partyOrPartyRole": {
                "id": "CUST3",
                "name": "Shawn Water",
                "@type": "PartyRef",
                "@referredType": "Individual"
              }
            },
            "value": {
              "amount": 0.7,
              "units": "GB"
            },
            "@type": "ConsumptionSummary"
          },
          {
            "id": "CT5",
            "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT5",
            "counterType": "used",
            "level": "detailed by user",
            "consumptionPeriod": {
              "endDateTime": "2021-06-01T16:04:09.084Z",
              "startDateTime": "2021-06-18T16:04:09.084Z"
            },
            "user": {
              "role": "Shared data member",
              "@type": "RelatedPartyRefOrPartyRoleRef",
              "partyOrPartyRole": {
                "id": "CUST1",
                "name": "John Water",
                "@type": "PartyRef",
                "@referredType": "Individual"
              }
            },
```


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```
            "value": {
              "amount": 1.5,
              "units": "GB"
            },
            "@type": "ConsumptionSummary"
          },
          {
            "id": "CT6",
            "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT6",
            "counterType": "remaining",
            "level": "detailed by user",
            "consumptionPeriod": {
              "endDateTime": "2021-06-01T16:04:09.084Z",
              "startDateTime": "2021-06-18T16:04:09.084Z"
            },
            "user": {
              "role": "Shared data member",
              "@type": "RelatedPartyRefOrPartyRoleRef",
              "partyOrPartyRole": {
                "id": "CUST2",
                "name": "Helen Water",
                "@type": "PartyRef",
                "@referredType": "Individual"
              }
            },
            "value": {
              "amount": 1,
              "units": "GB"
            },
            "@type": "ConsumptionSummary"
          },
          {
            "id": "CT7",
            "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT7",
            "counterType": "remaining",
            "level": "detailed by user",
            "consumptionPeriod": {
              "endDateTime": "2021-06-01T16:04:09.084Z",
              "startDateTime": "2021-06-18T16:04:09.084Z"
            },
            "user": {
              "role": "Shared data member",
              "@type": "RelatedPartyRefOrPartyRoleRef",
              "partyOrPartyRole": {
                "id": "CUST3",
                "name": "Shawn Water",
                "@type": "PartyRef",
                "@referredType": "Individual"
              }
            },
            "value": {
              "amount": 1.3,
              "units": "GB"
            },
            "@type": "ConsumptionSummary"
          },
          {
            "id": "CT8",
            "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT8",
            "counterType": "remaining",
```


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```
              "level": "detailed by user",
              "consumptionPeriod": {
                "endDateTime": "2021-06-01T16:04:09.084Z",
                "startDateTime": "2021-06-18T16:04:09.084Z"
              },
              "user": {
                "role": "Shared data member",
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "partyOrPartyRole": {
                  "id": "CUST1",
                  "name": "John Water",
                  "@type": "PartyRef",
                  "@referredType": "Individual"
                }
              },
              "value": {
                "amount": 4.5,
                "units": "GB"
              },
              "@type": "ConsumptionSummary"
            }
          ],
          "product": [
            {
              "id": "PR2",
              "href": "https://mycsp.com:8080/tmf-
api/productInventory/v5/product/PR2",
              "name": "RED Shared Data 10",
              "@type": "ProductRef"
            }
          ],
          "relatedParty": [
            {
              "role": "Customer",
              "@type": "RelatedPartyRefOrPartyRoleRef",
              "partyOrPartyRole": {
                "id": "3426",
                "href": "https://mycsp.com:8080/tmf-
api/customerManagement/v5/customer/3426",
                "name": "John Doe",
                "@type": "PartyRoleRef",
                "@referredType": "Customer"
              }
            }
          ],
          "@type": "Bucket"
        }
      ],
      "@type": "UsageConsumptionReport"
    }
  },
  "reportingSystem": {
    "id": "694",
    "name": "APP-764",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "621",
    "name": "APP-625",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
```


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```
  "@baseType": "Event",
  "@type": "UsageConsumptionReportCreateEvent"
}
```

Delete Event

Message example for UsageConsumptionReportDeleteEvent event

```
Content-Type: application/json
{
  "correlationId": "6cd1acc3-f745",
  "description": "UsageConsumptionReport DeleteEvent illustration",
  "domain": "Commercial",
  "eventId": "410d-b31d-62d7dcd302e7",
  "eventTime": "2022-08-25T12:25:47.711Z",
  "eventType": "UsageConsumptionReportDeleteEvent",
  "priority": "4",
  "timeOccurred": "2022-08-25T12:25:45.717Z",
  "title": "UsageConsumptionReportDeleteEvent",
  "event": {
    "usageConsumptionReport": {
      "id": "UC1",
      "creationDate": "2021-06-18T16:04:09.084Z",
      "lastUpdate": "2021-06-18T16:04:09.084Z",
      "bucket": [
        {
          "id": "BK1",
          "href": "https://mycsp.com:8080/tmf-api/balanceManagement/v5/bucket/BK1",
          "description": "Bucket related to international calls",
          "isShared": false,
          "name": "RED Voice Plus Bucket - International calls",
          "remainingValue": {
            "amount": 2000,
            "units": "minutes"
          },
          "remainingValueName": "2000 minutes",
          "usageType": "voice international",
          "bucketCounter": [
            {
              "id": "CT1",
              "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT1",
              "counterType": "used",
              "level": "global",
              "consumptionPeriod": {
                "endDateTime": "2021-06-01T16:04:09.084Z",
                "startDateTime": "2021-06-18T16:04:09.084Z"
              },
              "value": {
                "amount": 1000,
                "units": "minutes"
              },
              "@type": "BucketCounter"
            }
          ],
          "bucketRelationship": [
            {
              "relationshipType": "aggregated",
              "bucket": {
                "id": "BK2",
```

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```
                "href": "https://mycsp.com:8080/tmf-
api/usageConsumption/v5/bucket/BK2",
                "@type": "BucketRef"
              },
              "@type": "BucketRelationship"
            }
          ],
          "product": [
            {
              "id": "PR1",
              "href": "https://mycsp.com:8080/tmf-
api/productInventory/v5/product/PR1",
              "name": "RED Voice Plus",
              "@type": "ProductRef"
            }
          ],
          "relatedParty": [
            {
              "role": "Customer",
              "@type": "RelatedPartyRefOrPartyRoleRef",
              "partyOrPartyRole": {
                "id": "3426",
                "href": "https://mycsp.com:8080/tmf-
api/customerManagement/v5/customer/3426",
                "name": "John Doe",
                "@type": "PartyRoleRef",
                "@referredType": "Customer"
              }
            }
          ],
          "@type": "Bucket"
        },
        {
          "id": "BK2",
          "href": "https://mycsp.com:8080/tmf-api/balanceManagement/v5/bucket/BK2",
          "description": "Bucket related to national calls",
          "isShared": false,
          "name": "RED Voice Plus Bucket - National calls",
          "remainingValue": {
            "amount": 4000,
            "units": "minutes"
          },
          "remainingValueName": "4000 minutes",
          "usageType": "voice national",
          "bucketCounter": [
            {
              "id": "CT2",
              "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT2",
              "counterType": "used",
              "level": "global",
              "consumptionPeriod": {
                "endDateTime": "2021-06-01T16:04:09.084Z",
                "startDateTime": "2021-06-18T16:04:09.084Z"
              },
              "value": {
                "amount": 1000,
                "units": "minutes"
              },
              "@type": "ConsumptionSummary"
            }
          ],
          "bucketRelationship": [
```


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```
            {
              "relationshipType": "aggregated",
              "bucket": {
                "id": "BK1",
                "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/bucket/BK2",
                "@type": "BucketRef"
              },
              "@type": "BucketRelationship"
            }
          ],
          "product": [
            {
              "id": "PR1",
              "href": "https://mycsp.com:8080/tmf-
api/productInventory/v5/product/PR1",
              "name": "RED Voice Plus",
              "@type": "ProductRef"
            }
          ],
          "relatedParty": [
            {
              "role": "Customer",
              "@type": "RelatedPartyRefOrPartyRoleRef",
              "partyOrPartyRole": {
                "id": "3426",
                "href": "https://mycsp.com:8080/tmf-
api/customerManagement/v5/customer/3426",
                "name": "John Doe",
                "@type": "PartyRoleRef",
                "@referredType": "Customer"
              }
            }
          ],
          "@type": "Bucket"
        },
        {
          "id": "BK3",
          "href": "https://mycsp.com:8080/tmf-api/balanceManagement/v5/bucket/BK3",
          "description": "Bucket related to data",
          "isShared": true,
          "name": "RED Shared Data 10 - Data",
          "usageType": "data",
          "bucketCounter": [
            {
              "id": "CT3",
              "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT3",
              "counterType": "used",
              "level": "detailed by user",
              "consumptionPeriod": {
                "endDateTime": "2021-06-01T16:04:09.084Z",
                "startDateTime": "2021-06-18T16:04:09.084Z"
              },
              "user": {
                "role": "Shared data member",
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "partyOrPartyRole": {
                  "id": "CUST2",
                  "name": "Helen Water",
                  "@type": "PartyRef",
                  "@referredType": "Individual"
                }
```


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```
            },
            "value": {
              "amount": 1,
              "units": "GB"
            },
            "@type": "ConsumptionSummary"
          },
          {
            "id": "CT4",
            "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT4",
            "counterType": "used",
            "level": "detailed by user",
            "consumptionPeriod": {
              "endDateTime": "2021-06-01T16:04:09.084Z",
              "startDateTime": "2021-06-18T16:04:09.084Z"
            },
            "user": {
              "role": "Shared data member",
              "@type": "RelatedPartyRefOrPartyRoleRef",
              "partyOrPartyRole": {
                "id": "CUST3",
                "name": "Shawn Water",
                "@type": "PartyRef",
                "@referredType": "Individual"
              }
            },
            "value": {
              "amount": 0.7,
              "units": "GB"
            },
            "@type": "ConsumptionSummary"
          },
          {
            "id": "CT5",
            "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT5",
            "counterType": "used",
            "level": "detailed by user",
            "consumptionPeriod": {
              "endDateTime": "2021-06-01T16:04:09.084Z",
              "startDateTime": "2021-06-18T16:04:09.084Z"
            },
            "user": {
              "role": "Shared data member",
              "@type": "RelatedPartyRefOrPartyRoleRef",
              "partyOrPartyRole": {
                "id": "CUST1",
                "name": "John Water",
                "@type": "PartyRef",
                "@referredType": "Individual"
              }
            },
            "value": {
              "amount": 1.5,
              "units": "GB"
            },
            "@type": "ConsumptionSummary"
          },
          {
            "id": "CT6",
            "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT6",
```


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```
              "counterType": "remaining",
              "level": "detailed by user",
              "consumptionPeriod": {
                "endDateTime": "2021-06-01T16:04:09.084Z",
                "startDateTime": "2021-06-18T16:04:09.084Z"
              },
              "user": {
                "role": "Shared data member",
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "partyOrPartyRole": {
                  "id": "CUST2",
                  "name": "Helen Water",
                  "@type": "PartyRef",
                  "@referredType": "Individual"
                }
              },
              "value": {
                "amount": 1,
                "units": "GB"
              },
              "@type": "ConsumptionSummary"
            },
            {
              "id": "CT7",
              "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT7",
              "counterType": "remaining",
              "level": "detailed by user",
              "consumptionPeriod": {
                "endDateTime": "2021-06-01T16:04:09.084Z",
                "startDateTime": "2021-06-18T16:04:09.084Z"
              },
              "user": {
                "role": "Shared data member",
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "partyOrPartyRole": {
                  "id": "CUST3",
                  "name": "Shawn Water",
                  "@type": "PartyRef",
                  "@referredType": "Individual"
                }
              },
              "value": {
                "amount": 1.3,
                "units": "GB"
              },
              "@type": "ConsumptionSummary"
            },
            {
              "id": "CT8",
              "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT8",
              "counterType": "remaining",
              "level": "detailed by user",
              "consumptionPeriod": {
                "endDateTime": "2021-06-01T16:04:09.084Z",
                "startDateTime": "2021-06-18T16:04:09.084Z"
              },
              "user": {
                "role": "Shared data member",
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "partyOrPartyRole": {
                  "id": "CUST1",
```


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```
                  "name": "John Water",
                  "@type": "PartyRef",
                  "@referredType": "Individual"
                }
              },
              "value": {
                "amount": 4.5,
                "units": "GB"
              },
              "@type": "ConsumptionSummary"
            }
          ],
          "product": [
            {
              "id": "PR2",
              "href": "https://mycsp.com:8080/tmf-
api/productInventory/v5/product/PR2",
              "name": "RED Shared Data 10",
              "@type": "ProductRef"
            }
          ],
          "relatedParty": [
            {
              "role": "Customer",
              "@type": "RelatedPartyRefOrPartyRoleRef",
              "partyOrPartyRole": {
                "id": "3426",
                "href": "https://mycsp.com:8080/tmf-
api/customerManagement/v5/customer/3426",
                "name": "John Doe",
                "@type": "PartyRoleRef",
                "@referredType": "Customer"
              }
            }
          ],
          "@type": "Bucket"
        }
      ],
      "@type": "UsageConsumptionReport"
    }
  },
  "reportingSystem": {
    "id": "694",
    "name": "APP-764",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "621",
    "name": "APP-625",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "UsageConsumptionReportDeleteEvent"
}
```


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API OPERATIONS

Remember the following Uniform Contract:

Operation on Entities	Uniform API Operation	Description
Query Entities	GET Resource	GET must be used to retrieve a representation of a resource.
Create Entity	POST Resource	POST must be used to create a new resource
Partial Update of an Entity	PATCH Resource	PATCH must be used to partially update a resource
Remove an Entity	DELETE Resource	DELETE must be used to remove a resource
Execute an Action on an Entity	POST on TASK Resource	POST must be used to execute Task Resources
Other Request Methods	POST on TASK Resource	GET and POST must not be used to tunnel other request methods.

Filtering and attribute selection rules are described in the TMF REST Design Guidelines.

Notifications are also described in a subsequent section.

Operations on QueryUsageConsumption

List or find QueryUsageConsumption objects

`GET /queryUsageConsumption?fields=…`

Description

This operation list QueryUsageConsumption entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving QueryUsageConsumption resources.

Request

```
GET /queryUsageConsumption
Content-Type: application/json
```

Response

```
200
[
```

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```
  {
    "id": "7655",
    "href": "https://mycsp.com:8080/tmf-
api/usageConsumption/v5/queryUsageConsumption/7655",
    "relatedParty": [
      {
        "role": "Customer",
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "partyOrPartyRole": {
          "id": "3426",
          "href": "https://mycsp.com:8080/tmf-
api/customerManagement/v5/customer/3426",
          "name": "John Doe",
          "@type": "PartyRoleRef",
          "@referredType": "Customer"
        }
      }
    ],
    "searchCriteria": {
      "bucket": [
        {
          "usageType": "voice national",
          "@type": "Bucket"
        },
        {
          "usageType": "voice international",
          "@type": "Bucket"
        },
        {
          "usageType": "data",
          "@type": "Bucket"
        }
      ],
      "@type": "UsageConsumptionReport"
    },
    "@type": "QueryUsageConsumption"
  },
  {
    "id": "3853",
    "href": "https://mycsp.com:8080/tmf-
api/usageConsumption/v5/queryUsageConsumption/3853",
    "relatedParty": [
      {
        "role": "Customer",
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "partyOrPartyRole": {
          "id": "6254",
          "href": "https://mycsp.com:8080/tmf-
api/customerManagement/v5/customer/6254",
          "name": "John Doe",
          "@type": "PartyRoleRef",
          "@referredType": "Customer"
        }
      }
    ],
    "searchCriteria": {
      "logicalResource": [
        {
          "id": "1754",
          "name": "Lara's phone number",
          "value": "963161043",
          "@referredType": "MSISDN",
          "@type": "LogicalResourceRef"
```


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```
        }
      ],
      "@type": "UsageConsumptionReport"
    },
    "@type": "QueryUsageConsumption"
  },
  {
    "id": "1732",
    "href": "https://mycsp.com:8080/tmf-
api/usageConsumption/v5/queryUsageConsumption/1732",
    "relatedParty": [
      {
        "role": "Customer",
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "partyOrPartyRole": {
          "id": "7211",
          "href": "https://mycsp.com:8080/tmf-
api/customerManagement/v5/customer/7211",
          "name": "John Doe",
          "@type": "PartyRoleRef",
          "@referredType": "Customer"
        }
      }
    ],
    "searchCriteria": {
      "partyAccount": [
        {
          "id": "BI12234",
          "@referredType": "BillingAccount",
          "@type": "PartyAccountRef"
        }
      ],
      "@type": "UsageConsumptionReport"
    },
    "@type": "QueryUsageConsumption"
  }
]
```


Retrieves a QueryUsageConsumption by ID

`GET /queryUsageConsumption/{id}?fields=…`

Description

This operation retrieves a QueryUsageConsumption entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving a QueryUsageConsumption resource

Request

```
GET /queryUsageConsumption/8cb645cb-279f-42f4-9d3b-72845bbfafbe
Content-Type: application/json
```

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Response


```
200
{
  "id": "7655",
  "href": "https://mycsp.com:8080/tmf-
api/usageConsumption/v5/queryUsageConsumption/7655",
  "relatedParty": [
    {
      "role": "Agent",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "id": "3426",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/3426",
        "name": "Bob Smith",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    }
  ],
  "searchCriteria": {
    "logicalResource": [
      {
        "id": "1754",
        "name": "Lara's phone number",
        "value": "963161043",
        "@referredType": "MSISDN",
        "@type": "LogicalResourceRef"
      }
    ],
    "@type": "UsageConsumptionReport"
  },
  "usageConsumption": [
    {
      "id": "UC1",
      "href": "https://mycsp.com:8080/tmf-
api/usageConsumption/v5/usageConsumptionReport/UC1",
      "creationDate": "2021-06-18T16:04:09.084Z",
      "lastUpdate": "2021-06-18T16:04:09.084Z",
      "product": [
        {
          "id": "PR1",
          "href": "https://mycsp.com:8080/tmf-api/productInventory/v5/product/PR1",
          "name": "Lara's mobile line",
          "consumptionSummary": [
            {
              "id": "CT1",
              "href": "https://mycsp.com:8080/tmf-
api/usageConsumption/v5/consumptionSummary/CT1",
              "counterType": "voice",
              "level": "global",
              "valueName": "Unlimited",
              "consumptionPeriod": {
                "endDateTime": "2021-06-01T16:04:09.084Z",
                "startDateTime": "2021-06-30T16:04:09.084Z"
              },
              "@type": "ConsumptionSummary"
            },
            {
              "id": "CT2",
              "href": "https://mycsp.com:8080/tmf-
```


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```
api/usageConsumption/v5/consumptionSummary/CT2",
              "counterType": "sms",
              "level": "global",
              "valueName": "Unlimited",
              "consumptionPeriod": {
                "endDateTime": "2021-06-01T16:04:09.084Z",
                "startDateTime": "2021-06-30T16:04:09.084Z"
              },
              "@type": "ConsumptionSummary"
            }
          ],
          "@type": "UsageConsumptionProductRef"
        }
      ],
      "bucket": [
        {
          "id": "BK1",
          "href": "https://mycsp.com:8080/tmf-api/balanceManagement/v5/bucket/BK1",
          "description": "Bucket related to MMS",
          "isShared": false,
          "name": "RED MMS Plus Bucket - MMS",
          "usageType": "mms",
          "bucketCounter": [
            {
              "id": "CT3",
              "href": "https://mycsp.com:8080/tmf-
api/usageConsumption/v5/consumptionSummary/CT1",
              "counterType": "used",
              "level": "global",
              "consumptionPeriod": {
                "endDateTime": "2021-06-01T16:04:09.084Z",
                "startDateTime": "2021-06-18T16:04:09.084Z"
              },
              "value": {
                "amount": 0,
                "units": "mms"
              },
              "@type": "ConsumptionSummary"
            }
          ],
          "product": [
            {
              "id": "PR1",
              "href": "https://mycsp.com:8080/tmf-
api/productInventory/v5/product/PR1",
              "name": "RED MMS Plus",
              "@type": "ProductRef"
            }
          ],
          "relatedParty": [
            {
              "role": "Customer",
              "@type": "RelatedPartyRefOrPartyRoleRef",
              "partyOrPartyRole": {
                "id": "3426",
                "href": "https://mycsp.com:8080/tmf-
api/customerManagement/v5/customer/3426",
                "name": "John Doe",
                "@type": "PartyRoleRef",
                "@referredType": "Customer"
              }
            }
          ],
```


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```
          "@type": "Bucket"
        },
        {
          "id": "BK2",
          "href": "https://mycsp.com:8080/tmf-api/balanceManagement/v5/bucket/BK1",
          "description": "Bucket related to data",
          "isShared": false,
          "name": "RED Data Plus Bucket - Data",
          "remainingValue": {
            "amount": 3,
            "units": "GB"
          },
          "remainingValueName": "3 GB",
          "usageType": "data",
          "bucketCounter": [
            {
              "id": "CT4",
              "href": "https://mycsp.com:8080/tmf-
api/usageConsumption/v5/consumptionSummary/CT1",
              "counterType": "used",
              "level": "global",
              "consumptionPeriod": {
                "endDateTime": "2021-06-01T16:04:09.084Z",
                "startDateTime": "2021-06-18T16:04:09.084Z"
              },
              "value": {
                "amount": 2,
                "units": "GB"
              },
              "@type": "ConsumptionSummary"
            }
          ],
          "product": [
            {
              "id": "PR2",
              "href": "https://mycsp.com:8080/tmf-
api/productInventory/v5/product/PR2",
              "name": "RED Data Plus",
              "@type": "ProductRef"
            }
          ],
          "relatedParty": [
            {
              "role": "Customer",
              "@type": "RelatedPartyRefOrPartyRoleRef",
              "partyOrPartyRole": {
                "id": "3426",
                "href": "https://mycsp.com:8080/tmf-
api/customerManagement/v5/customer/3426",
                "name": "John Doe",
                "@type": "PartyRoleRef",
                "@referredType": "Customer"
              }
            }
          ],
          "@type": "Bucket"
        }
      ],
      "@type": "UsageConsumptionReport"
    }
  ],
  "@type": "QueryUsageConsumption"
```


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```
}
```

Creates a QueryUsageConsumption

POST /queryUsageConsumption?fields=…

Description

This operation creates a QueryUsageConsumption entity.

Mandatory Attributes

Mandatory Attributes	Rule
@type	M

Usage samples

Here's an example of a request for creating a QueryUsageConsumption resource.

Request

```
POST /queryUsageConsumption
Content-Type: application/json
{
  "searchCriteria": {
    "bucket": [
      {
        "usageType": "voice national",
        "@type": "Bucket"
      },
      {
        "usageType": "voice international",
        "@type": "Bucket"
      },
      {
        "usageType": "data",
        "@type": "Bucket"
      }
    ],
    "@type": "UsageConsumptionReport"
  },
  "partyAccount": [
    {
      "id": "BI12234",
      "@type": "PartyAccountRef",
      "@referredType": "BillingAccount"
    }
  ],
  "relatedParty": [
    {
      "role": "Agent",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "id": "3426",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/3426",
        "name": "Bob Smith",
```

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```
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    }
  ],
  "@type": "QueryUsageConsumption"
}
```

Response

```
201
{
  "id": "7655",
  "href": "https://mycsp.com:8080/tmf-
api/usageConsumption/v5/queryUsageConsumption/7655",
  "state": "done",
  "relatedParty": [
    {
      "role": "Agent",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "id": "3426",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/individual/3426",
        "name": "Bob Smith",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    }
  ],
  "searchCriteria": {
    "bucket": [
      {
        "usageType": "voice national",
        "@type": "Bucket"
      },
      {
        "usageType": "voice international",
        "@type": "Bucket"
      },
      {
        "usageType": "data",
        "@type": "Bucket"
      }
    ],
    "@type": "UsageConsumptionReport"
  },
  "usageConsumption": [
    {
      "id": "UC1",
      "creationDate": "2021-06-18T16:04:09.084Z",
      "lastUpdate": "2021-06-18T16:04:09.084Z",
      "bucket": [
        {
          "id": "BK1",
          "href": "https://mycsp.com:8080/tmf-api/balanceManagement/v5/bucket/BK1",
          "description": "Bucket related to international calls",
          "isShared": false,
          "name": "RED Voice Plus Bucket - International calls",
          "remainingValue": {
            "amount": 2000,
```

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```
            "units": "minutes"
          },
          "remainingValueName": "2000 minutes",
          "usageType": "voice international",
          "bucketCounter": [
            {
              "id": "CT1",
              "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT1",
              "counterType": "used",
              "level": "global",
              "consumptionPeriod": {
                "endDateTime": "2021-06-01T16:04:09.084Z",
                "startDateTime": "2021-06-18T16:04:09.084Z"
              },
              "value": {
                "amount": 1000,
                "units": "minutes"
              },
              "@type": "BucketCounter"
            }
          ],
          "bucketRelationship": [
            {
              "relationshipType": "aggregated",
              "bucket": {
                "id": "BK2",
                "href": "https://mycsp.com:8080/tmf-
api/usageConsumption/v5/bucket/BK2",
                "@type": "BucketRef"
              },
              "@type": "BucketRelationship"
            }
          ],
          "product": [
            {
              "id": "PR1",
              "href": "https://mycsp.com:8080/tmf-
api/productInventory/v5/product/PR1",
              "name": "RED Voice Plus",
              "@type": "ProductRef"
            }
          ],
          "relatedParty": [
            {
              "role": "Customer",
              "@type": "RelatedPartyRefOrPartyRoleRef",
              "partyOrPartyRole": {
                "id": "3426",
                "href": "https://mycsp.com:8080/tmf-
api/customerManagement/v5/customer/3426",
                "name": "John Doe",
                "@type": "PartyRoleRef",
                "@referredType": "Customer"
              }
            }
          ],
          "@type": "Bucket"
        },
        {
          "id": "BK2",
          "href": "https://mycsp.com:8080/tmf-api/balanceManagement/v5/bucket/BK2",
          "description": "Bucket related to national calls",
```


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```
          "isShared": false,
          "name": "RED Voice Plus Bucket - National calls",
          "remainingValue": {
            "amount": 4000,
            "units": "minutes"
          },
          "remainingValueName": "4000 minutes",
          "usageType": "voice national",
          "bucketCounter": [
            {
              "id": "CT2",
              "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT2",
              "counterType": "used",
              "level": "global",
              "consumptionPeriod": {
                "endDateTime": "2021-06-01T16:04:09.084Z",
                "startDateTime": "2021-06-18T16:04:09.084Z"
              },
              "value": {
                "amount": 1000,
                "units": "minutes"
              },
              "@type": "ConsumptionSummary"
            }
          ],
          "bucketRelationship": [
            {
              "relationshipType": "aggregated",
              "bucket": {
                "id": "BK1",
                "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/bucket/BK2",
                "@type": "BucketRef"
              },
              "@type": "BucketRelationship"
            }
          ],
          "product": [
            {
              "id": "PR1",
              "href": "https://mycsp.com:8080/tmf-
api/productInventory/v5/product/PR1",
              "name": "RED Voice Plus",
              "@type": "ProductRef"
            }
          ],
          "relatedParty": [
            {
              "role": "Customer",
              "@type": "RelatedPartyRefOrPartyRoleRef",
              "partyOrPartyRole": {
                "id": "3426",
                "href": "https://mycsp.com:8080/tmf-
api/customerManagement/v5/customer/3426",
                "name": "John Doe",
                "@type": "PartyRoleRef",
                "@referredType": "Customer"
              }
            }
          ],
          "@type": "Bucket"
        },
```


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```
        {
          "id": "BK3",
          "href": "https://mycsp.com:8080/tmf-api/balanceManagement/v5/bucket/BK3",
          "description": "Bucket related to data",
          "isShared": true,
          "name": "RED Shared Data 10 - Data",
          "usageType": "data",
          "bucketCounter": [
            {
              "id": "CT3",
              "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT3",
              "counterType": "used",
              "level": "detailed by user",
              "consumptionPeriod": {
                "endDateTime": "2021-06-01T16:04:09.084Z",
                "startDateTime": "2021-06-18T16:04:09.084Z"
              },
              "user": {
                "role": "Shared data member",
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "partyOrPartyRole": {
                  "id": "CUST2",
                  "name": "Helen Water",
                  "@type": "PartyRef",
                  "@referredType": "Individual"
                }
              },
              "value": {
                "amount": 1,
                "units": "GB"
              },
              "@type": "ConsumptionSummary"
            },
            {
              "id": "CT4",
              "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT4",
              "counterType": "used",
              "level": "detailed by user",
              "consumptionPeriod": {
                "endDateTime": "2021-06-01T16:04:09.084Z",
                "startDateTime": "2021-06-18T16:04:09.084Z"
              },
              "user": {
                "role": "Shared data member",
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "partyOrPartyRole": {
                  "id": "CUST3",
                  "name": "Shawn Water",
                  "@type": "PartyRef",
                  "@referredType": "Individual"
                }
              },
              "value": {
                "amount": 0.7,
                "units": "GB"
              },
              "@type": "ConsumptionSummary"
            },
            {
              "id": "CT5",
              "href": "https://mycsp.com:8080/tmf-
```


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```
api/balanceManagement/v5/consumptionSummary/CT5",
              "counterType": "used",
              "level": "detailed by user",
              "consumptionPeriod": {
                "endDateTime": "2021-06-01T16:04:09.084Z",
                "startDateTime": "2021-06-18T16:04:09.084Z"
              },
              "user": {
                "role": "Shared data member",
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "partyOrPartyRole": {
                  "id": "CUST1",
                  "name": "John Water",
                  "@type": "PartyRef",
                  "@referredType": "Individual"
                }
              },
              "value": {
                "amount": 1.5,
                "units": "GB"
              },
              "@type": "ConsumptionSummary"
            },
            {
              "id": "CT6",
              "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT6",
              "counterType": "remaining",
              "level": "detailed by user",
              "consumptionPeriod": {
                "endDateTime": "2021-06-01T16:04:09.084Z",
                "startDateTime": "2021-06-18T16:04:09.084Z"
              },
              "user": {
                "role": "Shared data member",
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "partyOrPartyRole": {
                  "id": "CUST2",
                  "name": "Helen Water",
                  "@type": "PartyRef",
                  "@referredType": "Individual"
                }
              },
              "value": {
                "amount": 1,
                "units": "GB"
              },
              "@type": "ConsumptionSummary"
            },
            {
              "id": "CT7",
              "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT7",
              "counterType": "remaining",
              "level": "detailed by user",
              "consumptionPeriod": {
                "endDateTime": "2021-06-01T16:04:09.084Z",
                "startDateTime": "2021-06-18T16:04:09.084Z"
              },
              "user": {
                "role": "Shared data member",
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "partyOrPartyRole": {
```


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```
                  "id": "CUST3",
                  "name": "Shawn Water",
                  "@type": "PartyRef",
                  "@referredType": "Individual"
                }
              },
              "value": {
                "amount": 1.3,
                "units": "GB"
              },
              "@type": "ConsumptionSummary"
            },
            {
              "id": "CT8",
              "href": "https://mycsp.com:8080/tmf-
api/balanceManagement/v5/consumptionSummary/CT8",
              "counterType": "remaining",
              "level": "detailed by user",
              "consumptionPeriod": {
                "endDateTime": "2021-06-01T16:04:09.084Z",
                "startDateTime": "2021-06-18T16:04:09.084Z"
              },
              "user": {
                "role": "Shared data member",
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "partyOrPartyRole": {
                  "id": "CUST1",
                  "name": "John Water",
                  "@type": "PartyRef",
                  "@referredType": "Individual"
                }
              },
              "value": {
                "amount": 4.5,
                "units": "GB"
              },
              "@type": "ConsumptionSummary"
            }
          ],
          "product": [
            {
              "id": "PR2",
              "href": "https://mycsp.com:8080/tmf-
api/productInventory/v5/product/PR2",
              "name": "RED Shared Data 10",
              "@type": "ProductRef"
            }
          ],
          "relatedParty": [
            {
              "role": "Customer",
              "@type": "RelatedPartyRefOrPartyRoleRef",
              "partyOrPartyRole": {
                "id": "3426",
                "href": "https://mycsp.com:8080/tmf-
api/customerManagement/v5/customer/3426",
                "name": "John Doe",
                "@type": "PartyRoleRef",
                "@referredType": "Customer"
              }
            }
          ],
          "@type": "Bucket"
```


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```
        }
      ],
      "@type": "UsageConsumptionReport"
    }
  ],
  "@type": "QueryUsageConsumption"
}
```

Deletes a QueryUsageConsumption

`DELETE /queryUsageConsumption/{id}`

Description

This operation deletes a QueryUsageConsumption entity.

Usage samples

Here's an example of deleting a specific usage consumption query resource.

Request

```
DELETE /queryUsageConsumption/Q3USER11222
Content-Type: application/json
```

Response

```
204
```

Operations on UsageConsumptionReport

List or find UsageConsumptionReport objects

`GET /usageConsumptionReport?fields=…`

Description

This operation list UsageConsumptionReport entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving UsageConsumptionReport resources.

Request

```
GET /usageConsumptionReport?relatedParty.id=43fef998-a865-4gac-971d-795b1195daaa
Content-Type: application/json
```

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Response


```
200
[
  {
    "id": "ucr0001",
    "href": "https://serverRoot/tmf-
api/usageConsumption/v4/usageConsumptionReport/ucr0001",
    "name": "Usage consumption report ucr0001",
    "description": "Usage consumption Report for Kate smartphone",
    "effectiveDate": "2018-03-15T15:44:28Z",
    "product": [
      {
        "id": "product1",
        "href": "https://serverRoot/tmf-api/productInventory/v5/product/product1",
        "name": "Kate smartphone",
        "consumptionSummary": [
          {
            "counterType": "outOfBucket",
            "level": "global",
            "value": {
              "amount": 20,
              "units": "USD"
            },
            "valueName": "20 USD",
            "consumptionPeriod": {
              "startDateTime": "2018-03-01T00:00:00Z",
              "endDateTime": "2018-03-15T00:00:00Z"
            },
            "user": {
              "role": "user",
              "@type": "RelatedPartyRefOrPartyRoleRef",
              "partyOrPartyRole": {
                "id": "usr1",
                "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/usr1",
                "name": "Kate",
                "@type": "PartyRef",
                "@referredType": "Individual"
              }
            },
            "@type": "ConsumptionSummary"
          }
        ],
        "@type": "ProductRef"
      }
    ],
    "bucket": [
      {
        "id": "bkt001",
        "href": "https://serverRoot/tmf-api/balanceManagement/v4/bucket/ucr0001",
        "name": "Main offer - data",
        "usageType": "data",
        "isShared": false,
        "product": [
          {
            "id": "product1",
            "href": "https://serverRoot/tmf-
api/productInventory/v5/product/product1",
            "name": "Kate smartphone",
            "@type": "ProductRef"
```


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```
          }
        ],
        "remainingValue": {
          "amount": 1.8,
          "units": "Go"
        },
        "remainingValueName": "it remains 1.8 Go",
        "validFor": {
          "startDateTime": "2018-03-15T00:00:00Z",
          "endDateTime": "2018-03-30T00:00:00Z"
        },
        "bucketCounter": [
          {
            "counterType": "used",
            "level": "global",
            "value": {
              "amount": 1.2,
              "units": "Go"
            },
            "valueName": "1.2 Go used",
            "consumptionPeriod": {
              "startDateTime": "2018-03-01T00:00:00Z",
              "endDateTime": "2018-03-15T00:00:00Z"
            },
            "user": {
              "role": "user",
              "@type": "RelatedPartyRefOrPartyRoleRef",
              "partyOrPartyRole": {
                "id": "usr1",
                "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/usr1",
                "name": "Kate",
                "@type": "PartyRef",
                "@referredType": "Individual"
              }
            },
            "@type": "ConsumptionSummary"
          }
        ],
        "@type": "Bucket"
      },
      {
        "id": "bkt002",
        "href": "https://serverRoot/tmf-api/balanceManagement/v4/bucket/ucr0001",
        "name": "Main offer - national voice",
        "usageType": "national voice",
        "isShared": false,
        "product": [
          {
            "id": "product1",
            "href": "https://serverRoot/tmf-
api/productInventory/v5/product/product1",
            "name": "Kate smartphone",
            "@type": "ProductRef"
          }
        ],
        "remainingValue": {
          "amount": 80,
          "units": "mins"
        },
        "remainingValueName": "it remains 80 minutes",
        "validFor": {
          "startDateTime": "2018-03-15T00:00:00Z",
```


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```
          "endDateTime": "2018-03-30T00:00:00Z"
        },
        "bucketCounter": [
          {
            "counterType": "used",
            "level": "global",
            "value": {
              "amount": 40,
              "units": "mins"
            },
            "valueName": "40 mins used",
            "consumptionPeriod": {
              "startDateTime": "2018-03-01T00:00:00Z",
              "endDateTime": "2018-03-15T00:00:00Z"
            },
            "user": {
              "role": "user",
              "@type": "RelatedPartyRefOrPartyRoleRef",
              "partyOrPartyRole": {
                "id": "usr1",
                "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/usr1",
                "name": "Kate",
                "@type": "PartyRef",
                "@referredType": "Individual"
              }
            },
            "@type": "ConsumptionSummary"
          }
        ],
        "@type": "Bucket"
      },
      {
        "id": "bkt003",
        "href": "https://serverRoot/tmf-api/balanceManagement/v4/bucket/ucr0001",
        "name": "Main offer - sms",
        "usageType": "sms",
        "isShared": false,
        "product": [
          {
            "id": "product1",
            "href": "https://serverRoot/tmf-
api/productInventory/v5/product/product1",
            "name": "Kate smartphone",
            "@type": "ProductRef"
          }
        ],
        "remainingValue": {
          "amount": 95,
          "units": "sms"
        },
        "remainingValueName": "it remains 95 sms",
        "validFor": {
          "startDateTime": "2018-03-15T00:00:00Z",
          "endDateTime": "2018-03-30T00:00:00Z"
        },
        "bucketCounter": [
          {
            "counterType": "used",
            "level": "global",
            "value": {
              "amount": 25,
              "units": "sms"
```


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```
            },
            "valueName": "25 sms used",
            "consumptionPeriod": {
              "startDateTime": "2018-03-01T00:00:00Z",
              "endDateTime": "2018-03-15T00:00:00Z"
            },
            "user": {
              "role": "user",
              "@type": "RelatedPartyRefOrPartyRoleRef",
              "partyOrPartyRole": {
                "id": "usr1",
                "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/usr1",
                "name": "Kate",
                "@type": "PartyRef",
                "@referredType": "Individual"
              }
            },
            "@type": "ConsumptionSummary"
          }
        ],
        "@type": "Bucket"
      },
      {
        "id": "bkt004",
        "href": "https://serverRoot/tmf-api/balanceManagement/v4/bucket/ucr0001",
        "name": "Option Canada/USA - voice",
        "usageType": "voice",
        "isShared": false,
        "product": [
          {
            "id": "product1",
            "href": "https://serverRoot/tmf-
api/productInventory/v5/product/product1",
            "name": "Kate smartphone",
            "@type": "ProductRef"
          }
        ],
        "remainingValue": {
          "amount": 10,
          "units": "mins"
        },
        "remainingValueName": "it remains 10 minutes",
        "validFor": {
          "startDateTime": "2018-03-15T00:00:00Z",
          "endDateTime": "2018-03-30T00:00:00Z"
        },
        "bucketCounter": [
          {
            "counterType": "used",
            "level": "global",
            "value": {
              "amount": 20,
              "units": "mins"
            },
            "valueName": "20 mins used",
            "consumptionPeriod": {
              "startDateTime": "2018-03-01T00:00:00Z",
              "endDateTime": "2018-03-15T00:00:00Z"
            },
            "user": {
              "role": "user",
              "@type": "RelatedPartyRefOrPartyRoleRef",
```


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```
              "partyOrPartyRole": {
                "id": "usr1",
                "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/usr1",
                "name": "Kate",
                "@type": "PartyRef",
                "@referredType": "Individual"
              }
            },
            "@type": "ConsumptionSummary"
          }
        ],
        "@type": "Bucket"
      },
      {
        "id": "bkt005",
        "href": "https://serverRoot/tmf-api/balanceManagement/v4/bucket/ucr0001",
        "name": "Option Canada/USA - sms",
        "usageType": "sms",
        "isShared": false,
        "product": [
          {
            "id": "product1",
            "href": "https://serverRoot/tmf-
api/productInventory/v5/product/product1",
            "name": "Kate smartphone",
            "@type": "ProductRef"
          }
        ],
        "remainingValue": {
          "amount": 0,
          "units": "sms"
        },
        "remainingValueName": "Sms bucket exhausted",
        "validFor": {
          "startDateTime": "2018-03-15T00:00:00Z",
          "endDateTime": "2018-03-30T00:00:00Z"
        },
        "bucketCounter": [
          {
            "counterType": "used",
            "level": "global",
            "value": {
              "amount": 10,
              "units": "sms"
            },
            "valueName": "10 sms used",
            "consumptionPeriod": {
              "startDateTime": "2018-03-01T00:00:00Z",
              "endDateTime": "2018-03-15T00:00:00Z"
            },
            "user": {
              "role": "user",
              "@type": "RelatedPartyRefOrPartyRoleRef",
              "partyOrPartyRole": {
                "id": "usr1",
                "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/usr1",
                "name": "Kate",
                "@type": "PartyRef",
                "@referredType": "Individual"
              }
            },
```


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```
            "@type": "ConsumptionSummary"
          }
        ],
        "@type": "Bucket"
      }
    ],
    "@type": "UsageConsumptionReport"
  }
]
```

Retrieves a UsageConsumptionReport by ID

GET /usageConsumptionReport/{id}?fields=…

Description

This operation retrieves a UsageConsumptionReport entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving a UsageConsumptionReport resource

Request

```
GET /usageConsumptionReport/c3fef858-b864-4fac-971e-795b1195ded9
Content-Type: application/json
```

Response

```
200
{
  "id": "ucr0001",
  "href": "https://serverRoot/tmf-
api/usageConsumption/v4/usageConsumptionReport/ucr0001",
  "name": "Usage consumption report ucr0001",
  "description": "Usage consumption Report for Kate smartphone",
  "effectiveDate": "2018-03-15T15:44:28Z",
  "product": [
    {
      "id": "product1",
      "href": "https://serverRoot/tmf-api/productInventory/v5/product/product1",
      "name": "Kate smartphone",
      "consumptionSummary": [
        {
          "counterType": "outOfBucket",
          "level": "global",
          "value": {
            "amount": 20,
            "units": "USD"
          },
          "valueName": "20 USD",
          "consumptionPeriod": {
            "startDateTime": "2018-03-01T00:00:00Z",
```

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```
            "endDateTime": "2018-03-15T00:00:00Z"
          },
          "user": {
            "role": "user",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
              "id": "usr1",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/usr1",
              "name": "Kate",
              "@type": "PartyRef",
              "@referredType": "Individual"
            }
          },
          "@type": "ConsumptionSummary"
        }
      ],
      "@type": "ProductRef"
    }
  ],
  "bucket": [
    {
      "id": "bkt001",
      "href": "https://serverRoot/tmf-api/balanceManagement/v4/bucket/ucr0001",
      "name": "Main offer - data",
      "usageType": "data",
      "isShared": false,
      "product": [
        {
          "id": "product1",
          "href": "https://serverRoot/tmf-api/productInventory/v5/product/product1",
          "name": "Kate smartphone",
          "@type": "ProductRef"
        }
      ],
      "remainingValue": {
        "amount": 1.8,
        "units": "Go"
      },
      "remainingValueName": "it remains 1.8 Go",
      "validFor": {
        "startDateTime": "2018-03-15T00:00:00Z",
        "endDateTime": "2018-03-30T00:00:00Z"
      },
      "bucketCounter": [
        {
          "counterType": "used",
          "level": "global",
          "value": {
            "amount": 1.2,
            "units": "Go"
          },
          "valueName": "1.2 Go used",
          "consumptionPeriod": {
            "startDateTime": "2018-03-01T00:00:00Z",
            "endDateTime": "2018-03-15T00:00:00Z"
          },
          "user": {
            "role": "user",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
              "id": "usr1",
              "href": "https://mycsp.com:8080/tmf-
```


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```
api/partyManagement/v5/individual/usr1",
              "name": "Kate",
              "@type": "PartyRef",
              "@referredType": "Individual"
            }
          },
          "@type": "ConsumptionSummary"
        }
      ],
      "@type": "Bucket"
    },
    {
      "id": "bkt002",
      "href": "https://serverRoot/tmf-api/balanceManagement/v4/bucket/ucr0001",
      "name": "Main offer - national voice",
      "usageType": "national voice",
      "isShared": false,
      "product": [
        {
          "id": "product1",
          "href": "https://serverRoot/tmf-api/productInventory/v5/product/product1",
          "name": "Kate smartphone",
          "@type": "ProductRef"
        }
      ],
      "remainingValue": {
        "amount": 80,
        "units": "mins"
      },
      "remainingValueName": "it remains 80 minutes",
      "validFor": {
        "startDateTime": "2018-03-15T00:00:00Z",
        "endDateTime": "2018-03-30T00:00:00Z"
      },
      "bucketCounter": [
        {
          "counterType": "used",
          "level": "global",
          "value": {
            "amount": 40,
            "units": "mins"
          },
          "valueName": "40 mins used",
          "consumptionPeriod": {
            "startDateTime": "2018-03-01T00:00:00Z",
            "endDateTime": "2018-03-15T00:00:00Z"
          },
          "user": {
            "role": "user",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
              "id": "usr1",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/usr1",
              "name": "Kate",
              "@type": "PartyRef",
              "@referredType": "Individual"
            }
          },
          "@type": "ConsumptionSummary"
        }
      ],
      "@type": "Bucket"
```


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```
    },
    {
      "id": "bkt003",
      "href": "https://serverRoot/tmf-api/balanceManagement/v4/bucket/ucr0001",
      "name": "Main offer - sms",
      "usageType": "sms",
      "isShared": false,
      "product": [
        {
          "id": "product1",
          "href": "https://serverRoot/tmf-api/productInventory/v5/product/product1",
          "name": "Kate smartphone",
          "@type": "ProductRef"
        }
      ],
      "remainingValue": {
        "amount": 95,
        "units": "sms"
      },
      "remainingValueName": "it remains 95 sms",
      "validFor": {
        "startDateTime": "2018-03-15T00:00:00Z",
        "endDateTime": "2018-03-30T00:00:00Z"
      },
      "bucketCounter": [
        {
          "counterType": "used",
          "level": "global",
          "value": {
            "amount": 25,
            "units": "sms"
          },
          "valueName": "25 sms used",
          "consumptionPeriod": {
            "startDateTime": "2018-03-01T00:00:00Z",
            "endDateTime": "2018-03-15T00:00:00Z"
          },
          "user": {
            "role": "user",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
              "id": "usr1",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/usr1",
              "name": "Kate",
              "@type": "PartyRef",
              "@referredType": "Individual"
            }
          },
          "@type": "ConsumptionSummary"
        }
      ],
      "@type": "Bucket"
    },
    {
      "id": "bkt004",
      "href": "https://serverRoot/tmf-api/balanceManagement/v4/bucket/ucr0001",
      "name": "Option Canada/USA - voice",
      "usageType": "voice",
      "isShared": false,
      "product": [
        {
          "id": "product1",
```


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```
          "href": "https://serverRoot/tmf-api/productInventory/v5/product/product1",
          "name": "Kate smartphone",
          "@type": "ProductRef"
        }
      ],
      "remainingValue": {
        "amount": 10,
        "units": "mins"
      },
      "remainingValueName": "it remains 10 minutes",
      "validFor": {
        "startDateTime": "2018-03-15T00:00:00Z",
        "endDateTime": "2018-03-30T00:00:00Z"
      },
      "bucketCounter": [
        {
          "counterType": "used",
          "level": "global",
          "value": {
            "amount": 20,
            "units": "mins"
          },
          "valueName": "20 mins used",
          "consumptionPeriod": {
            "startDateTime": "2018-03-01T00:00:00Z",
            "endDateTime": "2018-03-15T00:00:00Z"
          },
          "user": {
            "role": "user",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
              "id": "usr1",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/usr1",
              "name": "Kate",
              "@type": "PartyRef",
              "@referredType": "Individual"
            }
          },
          "@type": "ConsumptionSummary"
        }
      ],
      "@type": "Bucket"
    },
    {
      "id": "bkt005",
      "href": "https://serverRoot/tmf-api/balanceManagement/v4/bucket/ucr0001",
      "name": "Option Canada/USA - sms",
      "usageType": "sms",
      "isShared": false,
      "product": [
        {
          "id": "product1",
          "href": "https://serverRoot/tmf-api/productInventory/v5/product/product1",
          "name": "Kate smartphone",
          "@type": "ProductRef"
        }
      ],
      "remainingValue": {
        "amount": 0,
        "units": "sms"
      },
      "remainingValueName": "Sms bucket exhausted",
```


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```
      "validFor": {
        "startDateTime": "2018-03-15T00:00:00Z",
        "endDateTime": "2018-03-30T00:00:00Z"
      },
      "bucketCounter": [
        {
          "counterType": "used",
          "level": "global",
          "value": {
            "amount": 10,
            "units": "sms"
          },
          "valueName": "10 sms used",
          "consumptionPeriod": {
            "startDateTime": "2018-03-01T00:00:00Z",
            "endDateTime": "2018-03-15T00:00:00Z"
          },
          "user": {
            "role": "user",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
              "id": "usr1",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/usr1",
              "name": "Kate",
              "@type": "PartyRef",
              "@referredType": "Individual"
            }
          },
          "@type": "ConsumptionSummary"
        }
      ],
      "@type": "Bucket"
    }
  ],
  "@type": "UsageConsumptionReport"
}
```


Creates a UsageConsumptionReport

`POST /usageConsumptionReport?fields=…`

Description

This operation creates a UsageConsumptionReport entity.

Mandatory Attributes

Mandatory Attributes	Rule
@type	M

Usage samples

Here's an example of a request for creating a UsageConsumptionReport resource.

Request

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```
POST /usageConsumptionReport
Content-Type: application/json
{
  "name": "Usage consumption report ucr0001",
  "description": "Usage consumption Report for Kate smartphone",
  "product": [
    {
      "id": "product1",
      "href": "https://serverRoot/tmf-api/productInventory/v5/product/product1",
      "name": "Kate smartphone",
      "consumptionSummary": [
        {
          "counterType": "outOfBucket",
          "level": "global",
          "value": {
            "amount": 20,
            "units": "USD"
          },
          "valueName": "20 USD",
          "consumptionPeriod": {
            "startDateTime": "2018-03-01T00:00:00Z",
            "endDateTime": "2018-03-15T00:00:00Z"
          },
          "user": {
            "role": "user",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
              "id": "usr1",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/usr1",
              "name": "Kate",
              "@type": "PartyRef",
              "@referredType": "Individual"
            }
          },
          "@type": "ConsumptionSummary"
        }
      ],
      "@type": "UsageConsumptionProductRef"
    }
  ],
  "bucket": [
    {
      "id": "bkt001",
      "href": "https://serverRoot/tmf-api/balanceManagement/v4/bucket/ucr0001",
      "name": "Main offer - data",
      "usageType": "data",
      "isShared": false,
      "product": [
        {
          "id": "product1",
          "href": "https://serverRoot/tmf-api/productInventory/v5/product/product1",
          "name": "Kate smartphone",
          "@type": "ProductRef"
        }
      ],
      "remainingValue": {
        "amount": 1.8,
        "units": "Go"
      },
      "remainingValueName": "it remains 1.8 Go",
      "validFor": {
```

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```
        "startDateTime": "2018-03-15T00:00:00Z",
        "endDateTime": "2018-03-30T00:00:00Z"
      },
      "bucketCounter": [
        {
          "counterType": "used",
          "level": "global",
          "value": {
            "amount": 1.2,
            "units": "Go"
          },
          "valueName": "1.2 Go used",
          "consumptionPeriod": {
            "startDateTime": "2018-03-01T00:00:00Z",
            "endDateTime": "2018-03-15T00:00:00Z"
          },
          "user": {
            "role": "user",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
              "id": "usr1",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/usr1",
              "name": "Kate",
              "@type": "PartyRef",
              "@referredType": "Individual"
            }
          },
          "@type": "ConsumptionSummary"
        }
      ],
      "@type": "Bucket"
    },
    {
      "id": "bkt002",
      "href": "https://serverRoot/tmf-api/balanceManagement/v4/bucket/ucr0001",
      "name": "Main offer - national voice",
      "usageType": "national voice",
      "isShared": false,
      "product": [
        {
          "id": "product1",
          "href": "https://serverRoot/tmf-api/productInventory/v5/product/product1",
          "name": "Kate smartphone",
          "@type": "ProductRef"
        }
      ],
      "remainingValue": {
        "amount": 80,
        "units": "mins"
      },
      "remainingValueName": "it remains 80 minutes",
      "validFor": {
        "startDateTime": "2018-03-15T00:00:00Z",
        "endDateTime": "2018-03-30T00:00:00Z"
      },
      "bucketCounter": [
        {
          "counterType": "used",
          "level": "global",
          "value": {
            "amount": 40,
            "units": "mins"
```


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```
          },
          "valueName": "40 mins used",
          "consumptionPeriod": {
            "startDateTime": "2018-03-01T00:00:00Z",
            "endDateTime": "2018-03-15T00:00:00Z"
          },
          "user": {
            "role": "user",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
              "id": "usr1",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/usr1",
              "name": "Kate",
              "@type": "PartyRef",
              "@referredType": "Individual"
            }
          },
          "@type": "ConsumptionSummary"
        }
      ],
      "@type": "Bucket"
    },
    {
      "id": "bkt003",
      "href": "https://serverRoot/tmf-api/balanceManagement/v4/bucket/ucr0001",
      "name": "Main offer - sms",
      "usageType": "sms",
      "isShared": false,
      "product": [
        {
          "id": "product1",
          "href": "https://serverRoot/tmf-api/productInventory/v5/product/product1",
          "name": "Kate smartphone",
          "@type": "ProductRef"
        }
      ],
      "remainingValue": {
        "amount": 95,
        "units": "sms"
      },
      "remainingValueName": "it remains 95 sms",
      "validFor": {
        "startDateTime": "2018-03-15T00:00:00Z",
        "endDateTime": "2018-03-30T00:00:00Z"
      },
      "bucketCounter": [
        {
          "counterType": "used",
          "level": "global",
          "value": {
            "amount": 25,
            "units": "sms"
          },
          "valueName": "25 sms used",
          "consumptionPeriod": {
            "startDateTime": "2018-03-01T00:00:00Z",
            "endDateTime": "2018-03-15T00:00:00Z"
          },
          "user": {
            "role": "user",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
```


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```
              "id": "usr1",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/usr1",
              "name": "Kate",
              "@type": "PartyRef",
              "@referredType": "Individual"
            }
          },
          "@type": "ConsumptionSummary"
        }
      ],
      "@type": "Bucket"
    },
    {
      "id": "bkt004",
      "href": "https://serverRoot/tmf-api/balanceManagement/v4/bucket/ucr0001",
      "name": "Option Canada/USA - voice",
      "usageType": "voice",
      "isShared": false,
      "product": [
        {
          "id": "product1",
          "href": "https://serverRoot/tmf-api/productInventory/v5/product/product1",
          "name": "Kate smartphone",
          "@type": "ProductRef"
        }
      ],
      "remainingValue": {
        "amount": 10,
        "units": "mins"
      },
      "remainingValueName": "it remains 10 minutes",
      "validFor": {
        "startDateTime": "2018-03-15T00:00:00Z",
        "endDateTime": "2018-03-30T00:00:00Z"
      },
      "bucketCounter": [
        {
          "counterType": "used",
          "level": "global",
          "value": {
            "amount": 20,
            "units": "mins"
          },
          "valueName": "20 mins used",
          "consumptionPeriod": {
            "startDateTime": "2018-03-01T00:00:00Z",
            "endDateTime": "2018-03-15T00:00:00Z"
          },
          "user": {
            "role": "user",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
              "id": "usr1",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/usr1",
              "name": "Kate",
              "@type": "PartyRef",
              "@referredType": "Individual"
            }
          },
          "@type": "ConsumptionSummary"
        }
```


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```
      ],
      "@type": "Bucket"
    },
    {
      "id": "bkt005",
      "href": "https://serverRoot/tmf-api/balanceManagement/v4/bucket/ucr0001",
      "name": "Option Canada/USA - sms",
      "usageType": "sms",
      "isShared": false,
      "product": [
        {
          "id": "product1",
          "href": "https://serverRoot/tmf-api/productInventory/v5/product/product1",
          "name": "Kate smartphone",
          "@type": "ProductRef"
        }
      ],
      "remainingValue": {
        "amount": 0,
        "units": "sms"
      },
      "remainingValueName": "Sms bucket exhausted",
      "validFor": {
        "startDateTime": "2018-03-15T00:00:00Z",
        "endDateTime": "2018-03-30T00:00:00Z"
      },
      "bucketCounter": [
        {
          "counterType": "used",
          "level": "global",
          "value": {
            "amount": 10,
            "units": "sms"
          },
          "valueName": "10 sms used",
          "consumptionPeriod": {
            "startDateTime": "2018-03-01T00:00:00Z",
            "endDateTime": "2018-03-15T00:00:00Z"
          },
          "user": {
            "role": "user",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
              "id": "usr1",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/usr1",
              "name": "Kate",
              "@type": "PartyRef",
              "@referredType": "Individual"
            }
          },
          "@type": "ConsumptionSummary"
        }
      ],
      "@type": "Bucket"
    }
  ],
  "@type": "UsageConsumptionReport"
}
```


Response

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```
201
{
  "id": "ucr0001",
  "href": "https://serverRoot/tmf-
api/usageConsumption/v4/usageConsumptionReport/ucr0001",
  "name": "Usage consumption report ucr0001",
  "description": "Usage consumption Report for Kate smartphone",
  "creationDate": "2018-03-15T15:44:28Z",
  "product": [
    {
      "id": "product1",
      "href": "https://serverRoot/tmf-api/productInventory/v5/product/product1",
      "name": "Kate smartphone",
      "consumptionSummary": [
        {
          "counterType": "outOfBucket",
          "level": "global",
          "value": {
            "amount": 20,
            "units": "USD"
          },
          "valueName": "20 USD",
          "consumptionPeriod": {
            "startDateTime": "2018-03-01T00:00:00Z",
            "endDateTime": "2018-03-15T00:00:00Z"
          },
          "user": {
            "role": "user",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
              "id": "usr1",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/usr1",
              "name": "Kate",
              "@type": "PartyRef",
              "@referredType": "Individual"
            }
          },
          "@type": "ConsumptionSummary"
        }
      ],
      "@type": "UsageConsumptionProductRef"
    }
  ],
  "bucket": [
    {
      "id": "bkt001",
      "href": "https://serverRoot/tmf-api/balanceManagement/v4/bucket/ucr0001",
      "name": "Main offer - data",
      "usageType": "data",
      "isShared": false,
      "product": [
        {
          "id": "product1",
          "href": "https://serverRoot/tmf-api/productInventory/v5/product/product1",
          "name": "Kate smartphone",
          "@type": "ProductRef"
        }
      ],
      "remainingValue": {
        "amount": 1.8,
        "units": "Go"
```


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```
      },
      "remainingValueName": "it remains 1.8 Go",
      "validFor": {
        "startDateTime": "2018-03-15T00:00:00Z",
        "endDateTime": "2018-03-30T00:00:00Z"
      },
      "bucketCounter": [
        {
          "counterType": "used",
          "level": "global",
          "value": {
            "amount": 1.2,
            "units": "Go"
          },
          "valueName": "1.2 Go used",
          "consumptionPeriod": {
            "startDateTime": "2018-03-01T00:00:00Z",
            "endDateTime": "2018-03-15T00:00:00Z"
          },
          "user": {
            "role": "user",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
              "id": "usr1",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/usr1",
              "name": "Kate",
              "@type": "PartyRef",
              "@referredType": "Individual"
            }
          },
          "@type": "ConsumptionSummary"
        }
      ],
      "@type": "Bucket"
    },
    {
      "id": "bkt002",
      "href": "https://serverRoot/tmf-api/balanceManagement/v4/bucket/ucr0001",
      "name": "Main offer - national voice",
      "usageType": "national voice",
      "isShared": false,
      "product": [
        {
          "id": "product1",
          "href": "https://serverRoot/tmf-api/productInventory/v5/product/product1",
          "name": "Kate smartphone",
          "@type": "ProductRef"
        }
      ],
      "remainingValue": {
        "amount": 80,
        "units": "mins"
      },
      "remainingValueName": "it remains 80 minutes",
      "validFor": {
        "startDateTime": "2018-03-15T00:00:00Z",
        "endDateTime": "2018-03-30T00:00:00Z"
      },
      "bucketCounter": [
        {
          "counterType": "used",
          "level": "global",
```


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```
          "value": {
            "amount": 40,
            "units": "mins"
          },
          "valueName": "40 mins used",
          "consumptionPeriod": {
            "startDateTime": "2018-03-01T00:00:00Z",
            "endDateTime": "2018-03-15T00:00:00Z"
          },
          "user": {
            "role": "user",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
              "id": "usr1",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/usr1",
              "name": "Kate",
              "@type": "PartyRef",
              "@referredType": "Individual"
            }
          },
          "@type": "ConsumptionSummary"
        }
      ],
      "@type": "Bucket"
    },
    {
      "id": "bkt003",
      "href": "https://serverRoot/tmf-api/balanceManagement/v4/bucket/ucr0001",
      "name": "Main offer - sms",
      "usageType": "sms",
      "isShared": false,
      "product": [
        {
          "id": "product1",
          "href": "https://serverRoot/tmf-api/productInventory/v5/product/product1",
          "name": "Kate smartphone",
          "@type": "ProductRef"
        }
      ],
      "remainingValue": {
        "amount": 95,
        "units": "sms"
      },
      "remainingValueName": "it remains 95 sms",
      "validFor": {
        "startDateTime": "2018-03-15T00:00:00Z",
        "endDateTime": "2018-03-30T00:00:00Z"
      },
      "bucketCounter": [
        {
          "counterType": "used",
          "level": "global",
          "value": {
            "amount": 25,
            "units": "sms"
          },
          "valueName": "25 sms used",
          "consumptionPeriod": {
            "startDateTime": "2018-03-01T00:00:00Z",
            "endDateTime": "2018-03-15T00:00:00Z"
          },
          "user": {
```


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```
            "role": "user",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
              "id": "usr1",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/usr1",
              "name": "Kate",
              "@type": "PartyRef",
              "@referredType": "Individual"
            }
          },
          "@type": "ConsumptionSummary"
        }
      ],
      "@type": "Bucket"
    },
    {
      "id": "bkt004",
      "href": "https://serverRoot/tmf-api/balanceManagement/v4/bucket/ucr0001",
      "name": "Option Canada/USA - voice",
      "usageType": "voice",
      "isShared": false,
      "product": [
        {
          "id": "product1",
          "href": "https://serverRoot/tmf-api/productInventory/v5/product/product1",
          "name": "Kate smartphone",
          "@type": "ProductRef"
        }
      ],
      "remainingValue": {
        "amount": 10,
        "units": "mins"
      },
      "remainingValueName": "it remains 10 minutes",
      "validFor": {
        "startDateTime": "2018-03-15T00:00:00Z",
        "endDateTime": "2018-03-30T00:00:00Z"
      },
      "bucketCounter": [
        {
          "counterType": "used",
          "level": "global",
          "value": {
            "amount": 20,
            "units": "mins"
          },
          "valueName": "20 mins used",
          "consumptionPeriod": {
            "startDateTime": "2018-03-01T00:00:00Z",
            "endDateTime": "2018-03-15T00:00:00Z"
          },
          "user": {
            "role": "user",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
              "id": "usr1",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/usr1",
              "name": "Kate",
              "@type": "PartyRef",
              "@referredType": "Individual"
            }
```


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```
          },
          "@type": "ConsumptionSummary"
        }
      ],
      "@type": "Bucket"
    },
    {
      "id": "bkt005",
      "href": "https://serverRoot/tmf-api/balanceManagement/v4/bucket/ucr0001",
      "name": "Option Canada/USA - sms",
      "usageType": "sms",
      "isShared": false,
      "product": [
        {
          "id": "product1",
          "href": "https://serverRoot/tmf-api/productInventory/v5/product/product1",
          "name": "Kate smartphone",
          "@type": "ProductRef"
        }
      ],
      "remainingValue": {
        "amount": 0,
        "units": "sms"
      },
      "remainingValueName": "Sms bucket exhausted",
      "validFor": {
        "startDateTime": "2018-03-15T00:00:00Z",
        "endDateTime": "2018-03-30T00:00:00Z"
      },
      "bucketCounter": [
        {
          "counterType": "used",
          "level": "global",
          "value": {
            "amount": 10,
            "units": "sms"
          },
          "valueName": "10 sms used",
          "consumptionPeriod": {
            "startDateTime": "2018-03-01T00:00:00Z",
            "endDateTime": "2018-03-15T00:00:00Z"
          },
          "user": {
            "role": "user",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
              "id": "usr1",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/usr1",
              "name": "Kate",
              "@type": "PartyRef",
              "@referredType": "Individual"
            }
          },
          "@type": "ConsumptionSummary"
        }
      ],
      "@type": "Bucket"
    }
  ],
  "@type": "UsageConsumptionReport"
}
```


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Deletes a UsageConsumptionReport

`DELETE /usageConsumptionReport/{id}`

Description

This operation deletes a UsageConsumptionReport entity.

Usage samples

Here's an example of a request for deleting an usage consumption report resource.

Request

```
DELETE /usageConsumptionReport/ucr001
Content-Type: application/json
```

Response

```
204
```

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API NOTIFICATIONS

For every single of operation on the entities use the following templates and provide sample REST notification POST calls.

It is assumed that the Pub/Sub uses the Register and UnRegister mechanisms described in the REST Guidelines reproduced below.

Register listener

POST /hub

Description

Sets the communication endpoint address the service instance must use to deliver information about its health state, execution state, failures and metrics. Subsequent POST calls will be rejected by the service if it does not support multiple listeners. In this case DELETE /api/hub/{id} must be called before an endpoint can be created again.

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 409 if request is not successful.

Usage Samples

Here's an example of a request for registering a listener.

Request

```
POST /api/hub

Accept: application/json

{
    "callback": "http://in.listener.com"
}
```

Response

```
201

Content-Type: application/json

Location: /api/hub/42

{
    "id":"42",
    "callback": "http://in.listener.com",
    "query":""
}
```

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Unregister listener

DELETE /hub/{id}

Description

Clears the communication endpoint address that was set by creating the Hub..

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 404 if the resource is not found.

Usage Samples

Here's an example of a request for un-registering a listener.

Request

```
DELETE /api/hub/42

Accept: application/json
```

Response

```
204
```

Publish Event to listener

POST /client/listener

Description

Clears the communication endpoint address that was set by creating the Hub.

Provides to a registered listener the description of the event that was raised. The /client/listener url is the callback url passed when registering the listener.

Behavior

Returns HTTP/1.1 status code 201 if the service is able to set the configuration.

Usage Samples

Here's an example of a notification received by the listener. In this example "EVENT TYPE" should be replaced by one of the notification types supported by this API (see Notification resources Models section) and EVENT BODY refers to the data structure of the given notification type.

Request

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```
POST /client/listener

Accept: application/json

{
    "event": {
        EVENT BODY
    },
    "eventType": "EVENT_TYPE"
}
```

Response

```
201
```

For detailed examples on the general TM Forum notification mechanism, see the TMF REST Design Guidelines.

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Acknowledgements

Release History

Release Number	Date	Release led by:	Description
1.0	April 2017	Sophie Bouleau Orange sophie.bouleau@orange.com Luis Velarde Tazon, Telefonica luis.velardetazon@telefonica.com	First Release of the Document.
2.0	Nov 2018	Mariano Belaunde Orange Labs	Alignment with Guidelines 3.0
3.0	Jan 2019	Sophie Bouleau Orange sophie.bouleau@orange.com	Alignment with Guidelines 3.0 and with SID mapping. Changes from JIRA AP-1052
4.0.0	July 2021	Lara Silva Vodafone lara.silva3@vodafone.com	Introduction of the Bucket entity
5.0.0	Nov 2024	Rochana Macharoen AIS rochanam@ais.co.th	Alignment with Guidelines 5.0 Removed service consumption summary Removed resource consumption summary (used only for product identification purposes) Added UsageConsumptionReport managed entity

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